

S6

Correlation & Regression

Learning Objectives

In this chapter, we will

- (a) draw and use scatter diagram to judge if there is a plausible linear relationship between 2 variables;
- (b) calculate and interpret the product moment correlation coefficient and
 - use it as a measure of fit of a linear model to the scatter diagram
 - interpret it for values close to -1 , 0 and 1 ;
- (c) understand concepts of linear regression and method of least squares to find the equations of the regression lines;
- (d) understand concepts of interpolation and extrapolation;
- (e) use of appropriate regression lines to make prediction, estimation and explain how well the situation is modelled by the regression model;
- (f) use of suitable transformations such as square, reciprocal or a logarithmic transformation to achieve linearity.

= Indicates section/example with accompanying video explanation.

(Use the link in the softcopy for access.)

6.1 # Introduction

In the previous chapters, we have been studying data with one variable. In this chapter, we shall investigate data with two variables and their relationship. For example, if we can find a relationship between the midyear examination scores and year-end examination scores of students, we will be able to use the information to help us make statistical inferences about the two examination scores.

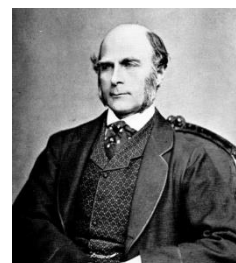
Examples of data with two variables include

- displacement of an object and its velocity,
- speed and time of a falling object,
- annual income level and education level of individuals,
- students' examination scores in Chemistry and Physics.

Data with two variables are known as **bivariate data** and they are usually expressed as **ordered pairs** (x, y) .

6.2 Bivariate Data

Studies of bivariate data first began in 1860s when Sir Francis Galton investigated the degree of resemblance between children and their parents. In a study carried out by Galton and his student, Karl Pearson, they measured the heights of 1078 fathers and the heights of their sons. In order to investigate the data, the first use of correlation and regression studies of data emerged.



Sir Francis Galton
(1822-1911)

Suppose we wish to examine the relationship between the midyear scores and final examination scores of students. We may want to find a model that can be used to predict the final examination score for a student having a known midyear examination score.

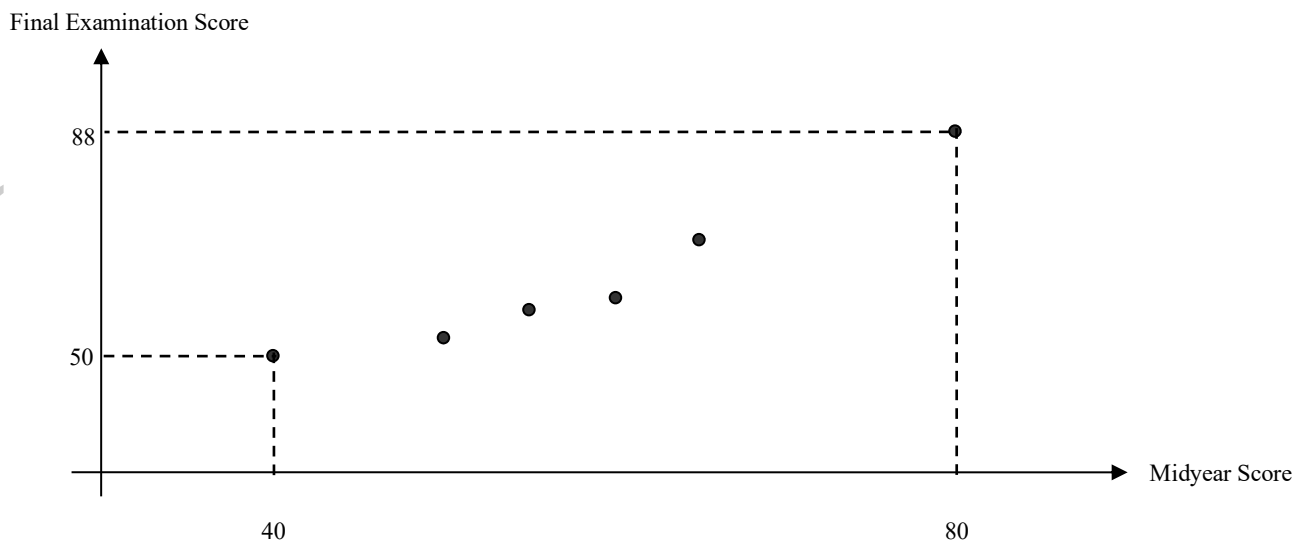
6.3 Scatter Diagram (or Scatter Plot)

A **scatter diagram** is obtained when each of the observed bivariate data (x_i, y_i) , $i = 1, 2, \dots, n$ is plotted on the Cartesian plane.

Each (x_i, y_i) on the scatter diagram represents a single data point. From the scatter diagram, we can judge visually if there is any relationship between x and y .

An example of a data set and its scatter diagram is shown below.

Student	A	B	C	D	E	F
Midyear score	40	50	55	60	65	80
Final examination score	50	53	58	60	70	88



Note

When we sketch a scatter diagram, it is important to

- 1) sketch according to scale, with correct relative positions and
- 2) indicate the range of the data set on both axes.

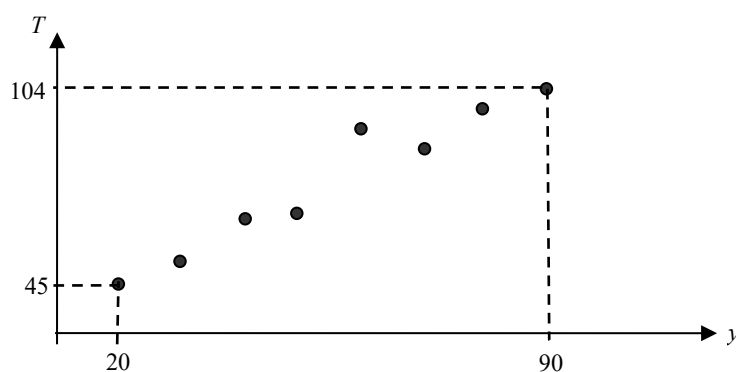
e.g. maximum and minimum Midyear and Final Examination Score labelled.

Example 1

The temperature, T , in degree Celsius ($^{\circ}\text{C}$) of the tyre of a car is measured when the car travels at different speed, v (kmh^{-1}). Eight sets of data are obtained. Sketch the scatter diagram for the data.

v	20	30	40	50	60	70	80	90
T	45	52	64	66	91	86	98	104

Solution



By GC,

- ① Create two new lists, L_1 and L_2 using the v and T values respectively.

Press **[stat]**, select **1:Edit...**, press **[enter]**. Key into L_1 and L_2 the values for v and T respectively.

NORMAL FLOAT AUTO REAL RADIAN MP					
L1	L2	L3	L4	L5	
20	45				
30	52				
40	64				
50	66				
60	91				
70	86				
80	98				
90	104				
-----	-----				

L2(8)=104

- ② To plot the data, press **[2nd][y=]** for [stat plot].
Press **[enter]** to select Plot 1.
Press **[enter]** to highlight On.
Under Type, choose first icon for Scatter plot.
To select L₁ for Xlist and L₂ for Ylist, press **[2nd][stat]** and select **1:L1** for L₁ and **2:L2** for L₂.
For 'Mark', select the desired style to represent the data points.

The screenshot shows the TI-84 Plus CSE graphing calculator screen. At the top, the status bar displays 'NORMAL FLOAT AUTO REAL RADIAN MP'. Below this, the 'Plot1' menu is active, showing 'Plot1 Plot2 Plot3'. The 'On Off' option is selected, with 'On' highlighted. Below 'On Off', the 'Type:' field shows a solid line icon. The 'Xlist:' field is set to 'L1', and the 'Ylist:' field is set to 'L2'. The 'Mark:' field shows a small square icon. The 'Color:' field is set to 'BLACK'.

- ③ To view the scatter diagram, press **[zoom]**, select **9:ZoomStat**, press **[enter]**.

To read the coordinates of each point, press **trace**.

Plot1:L1,L2

X=50 Y=66

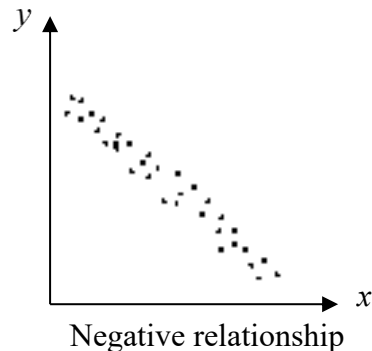
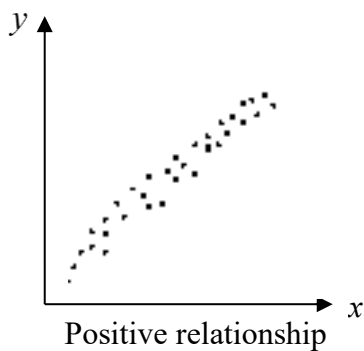
6.4 # Analysis of Scatter Diagrams

Scatter diagrams can reveal general patterns and relationships between two variables. We can comment on the

1. Direction

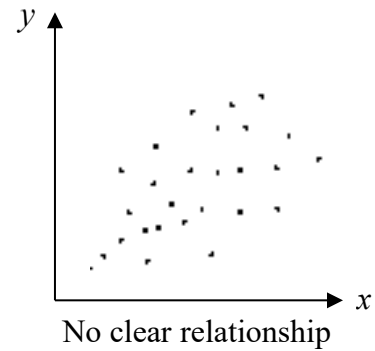
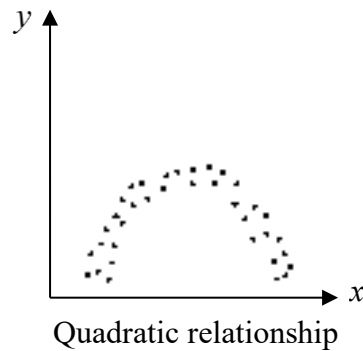
The two variables x and y can be positively correlated or negatively correlated.

- In general, if y increases as x increases, then x and y are **positively** correlated.
- In general, if y decreases as x increases, then x and y are **negatively** correlated.



2. Form

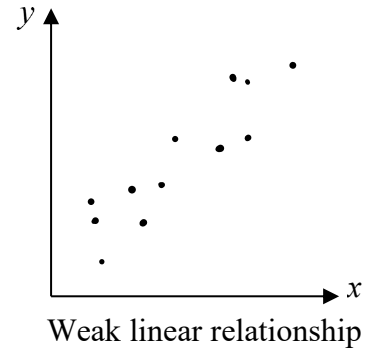
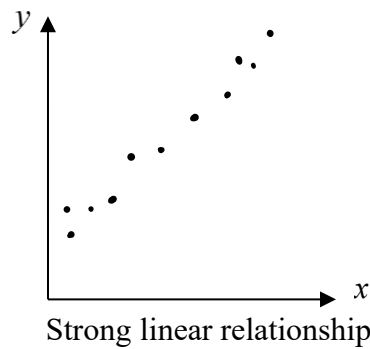
The form of a scatter diagram refers to the shape of the distribution of data points. The relationship may be linear or curvilinear and sometimes there is no clear relationship.



3. Strength

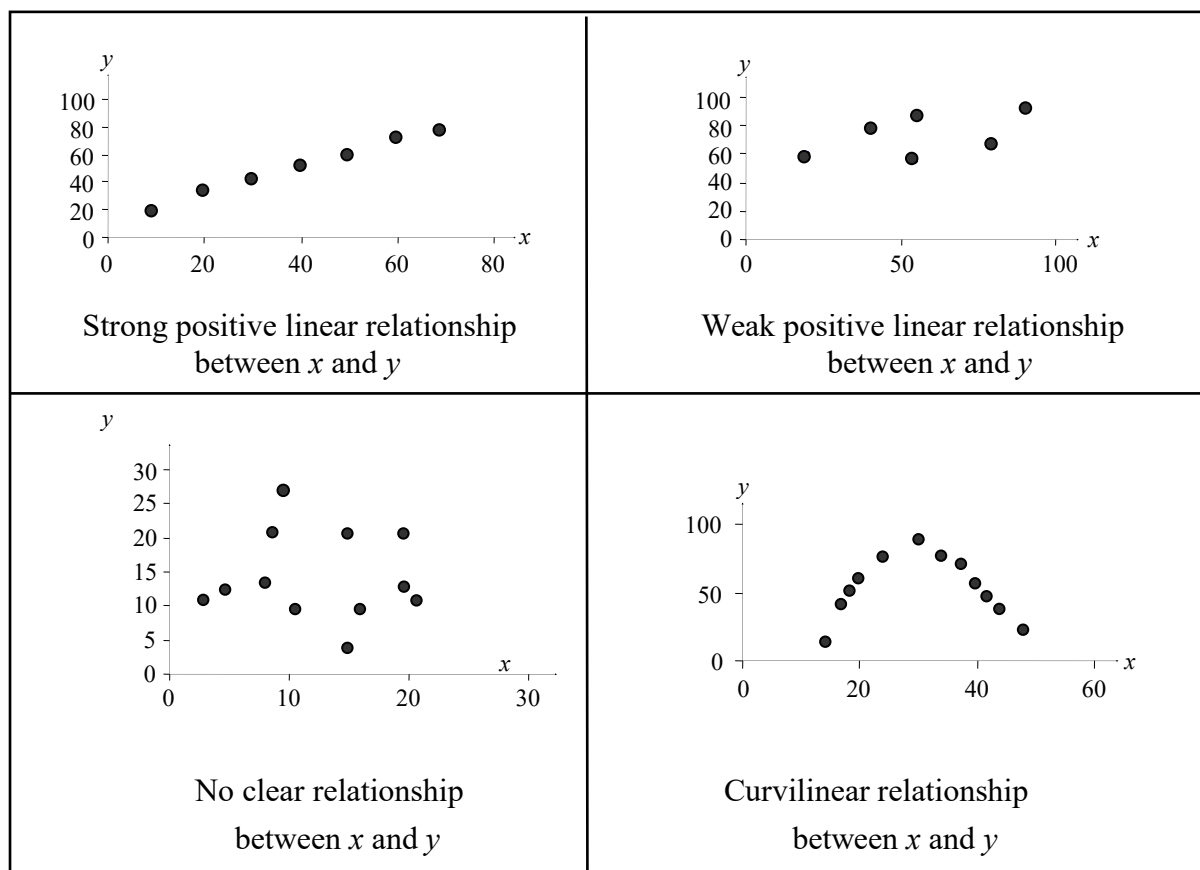
The strength of a scatter diagram describes how tightly clustered the points are to the underlying form.

- If the data points are tightly clustered to a straight line, there is a **strong** linear relation between x and y .
- If the data points are loosely clustered to a straight line, it may indicate a **weak** linear relation between x and y .



Example 2

Comment on the relationship between the two variables based on the scatter diagrams below.

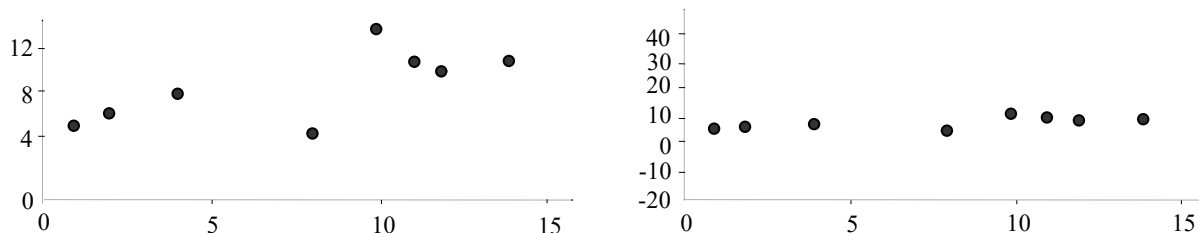


Note

Scatter diagrams can also give us visual evidence of **outliers** or suspicious observations. These data points are points in a sample that lie outside the overall pattern of a distribution.

6.5 Correlation

Interpretation of the relationship between the variables of sample data solely based on the scatter diagram is subjective. It can even be deceiving when different scales for the axes are used. The scatter diagrams below are plotted using the same set of data but on different scales for the y axis.



Since the scale of a scatter diagram can be manipulated, it may be more helpful to use a numerical approach to measure the strength of a relationship between two variables. **The product moment correlation coefficient, r , is used to measure the strength of a linear relationship.**

The formula for r with n data points, which can be found in the MF 27, is as follows:

Estimated Product Moment Correlation Coefficient

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}} = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sqrt{\left(\sum x^2 - \frac{(\sum x)^2}{n} \right) \left(\sum y^2 - \frac{(\sum y)^2}{n} \right)}}$$

Using $S_{xx} = \sum (x - \bar{x})^2 = \sum x^2 - \frac{(\sum x)^2}{n}$,

$$S_{yy} = \sum (y - \bar{y})^2 = \sum y^2 - \frac{(\sum y)^2}{n},$$

$$S_{xy} = \sum (x - \bar{x})(y - \bar{y}) = \sum xy - \frac{\sum x \sum y}{n},$$

we have $r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$.

Example 3

In a physical education class, the number of push-ups (x) and sit-ups (y) done by a sample of ten randomly chosen students were recorded and summarized as shown.

Student	1	2	3	4	5	6	7	8	9	10
x	27	22	15	35	30	52	35	55	40	40
y	30	26	25	42	38	40	32	54	50	43

$$\sum xy = 14257, \quad \sum x^2 = 13717, \quad \sum y^2 = 15298, \quad \sum x = 351, \quad \sum y = 380$$

Find the product moment correlation coefficient of the sample.

Solution

Method 1 (using MF27 formula)

$$r = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sqrt{\left(\sum x^2 - \frac{(\sum x)^2}{n} \right) \left(\sum y^2 - \frac{(\sum y)^2}{n} \right)}} = \frac{14257 - \frac{(351)(380)}{10}}{\sqrt{\left(13717 - \frac{(351)^2}{10} \right) \left(15298 - \frac{(380)^2}{10} \right)}} = 0.839 \text{ (3 s.f.)}$$

Method 2 (using GC since actual data is given)

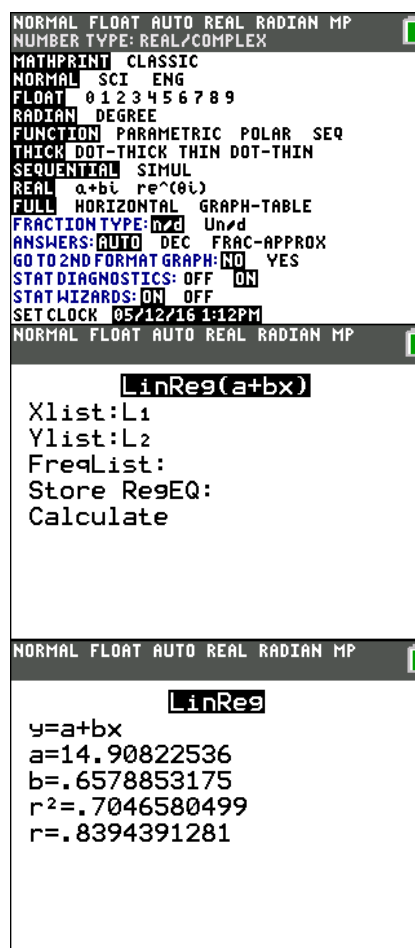
Since the actual data is given, we can use the graphic calculator to calculate the product moment correlation coefficient. However, **if it is the first data set the graphic calculator is used on this topic, we would need to perform the following steps to activate Stat Diagnostics.** This is to enable the computed value of r to be displayed. Subsequently, Stat Diagnostics will be switched on by default and we will not need to activate it manually.

- ① To turn on Stat Diagnostics for GC:

Press **[mode]** and scroll down to STAT DIAGNOSTICS, select ON, and press **[enter]**. Subsequently, Stat Diagnostics will be switched on by default and we will not need to activate it manually.

- ② Store data under L_1 (x) and L_2 (y). (Recall Example 1)
- ③ Press **[stat]** and select CALC. Select **8:LinReg(a+bx)** and press **[enter]**.
- ④ Key in L_1 under Xlist, then L_2 under Ylist and press Calculate.

From the results displayed, $r = 0.839$.



We will be looking at the significance of the other values that appear in the screenshot in the next section.

Note

1. For any sample data, $-1 \leq r \leq 1$.
2. The **sign** of the correlation coefficient indicates the direction of linear correlation.
When $r > 0$, the correlation between x and y is positive.
When $r < 0$, the correlation between x and y is negative.

3. The **magnitude** of r indicates the strength of the **linear correlation**.
Generally, the bigger the value of $|r|$, the stronger the linear relationship.

When $r = +1$, we have perfect positive linear correlation between x and y . All the points lie on a straight line with positive gradient.
When $r = -1$, we have perfect negative linear correlation between x and y . All the points lie on a straight line with negative gradient.
When $r = 0$, there is **no linear correlation** between x and y . **But it does not mean there is no relationship between the two variables.**
4. **A scatter diagram together with the product moment correlation coefficient should be used to determine if there is a linear relationship.** See Appendix A for an example of data sets with the same product moment correlation coefficient but different scatter diagrams.
5. A high correlation between two variables i.e. $|r| \approx 1$ **does not necessarily imply one variable has a causal effect on the other.**
6. In general,
 $0.8 < |r| < 1$ indicates strong linear correlation between the two variables.
 $0.5 < |r| < 0.8$ indicates moderately strong linear correlation between the two variables.
 $0 < |r| < 0.5$ indicates weak linear correlation between the two variables.
The above classification should only be treated as a guide.
7. The measure r has no units. It is **independent of the scale of measurement** of the variables. This means that r will not change if we add a constant or multiply a positive constant to **all** the values of a variable. i.e. if $s = a + bx$, $t = c + dy$, where a, b, c, d are constants, b and d both positive, then $r_{st} = r_{xy}$ (see Appendix B).

Example 4

The temperature, T , in degree Celsius ($^{\circ}\text{C}$) of the tyre of a car is measured when the car travels at different speed, v (kmh^{-1}). Eight sets of data are obtained.

v	20	30	40	50	60	70	80	90
T	45	52	64	66	91	86	98	104

- (i) Find the product moment correlation coefficient between v and T .
- (ii) Find the corresponding product moment correlation coefficient between v and F where F is the temperature of the tyre measured in degree Fahrenheit ($^{\circ}\text{F}$).
(Use the formula $F = \frac{9}{5}T + 32$)

Solution

(i) Using GC, $r = 0.975$ (to 3 s.f.)

- ① Store data under L_1 (x) and L_2 (y).
- ② Press **[stat]** and select CALC. Select **8:LinReg(a+bx)** and press **[enter]**.
- ③ Key in L_1 under Xlist, then L_2 under Ylist and press Calculate.

From the results displayed, $r = 0.975$ (to 3 s.f.)

NORMAL FLOAT AUTO REAL RADIAN MP					
L1	L2	L3	L4	L5	2
20	45	-----	-----	-----	
30	52				
40	64				
50	66				
60	91				
70	86				
80	98				
90	104				
-----	-----				
L2(8)=104					
NORMAL FLOAT AUTO REAL RADIAN MP					
LinReg					
y=a+bx					
a=27.69047619					
b=.8738095238					
r ² =.9506094419					
r=.9749920215					

(ii) Using GC, $r = 0.975$ (to 3 s.f.)

- ① Scroll up to the header L_3 and highlight by pressing **[enter]**.
- ② Key into the header L_3 the formula for F , which is $(9/5)L_2 + 32$, and press **[enter]**. Note that to enter L_2 , we press **[2nd][stat]** and select **2: L2**.
- ③ Press **[stat]** and select CALC. Select **8:LinReg(a+bx)** and press **[enter]**.
- ④ Key in L_1 under Xlist, then L_3 under Ylist and press Calculate.

From the results displayed, $r = 0.975$ (to 3 s.f.)

NORMAL FLOAT AUTO REAL RADIAN MP					
L1	L2	L3	L4	L5	3
20	45	-----	-----	-----	
30	52				
40	64				
50	66				
60	91				
70	86				
80	98				
90	104				
-----	-----				
L3=					

NORMAL FLOAT AUTO REAL RADIAN MP					
L1	L2	L3	L4	L5	3
20	45	113	-----	-----	
30	52	125.6			
40	64	147.2			
50	66	150.8			
60	91	195.8			
70	86	186.8			
80	98	208.4			
90	104	219.2			
-----	-----	-----			
L3={113,125.6,147.2,150.8,1					

NORMAL FLOAT AUTO REAL RADIAN MP					
LinReg					
y=a+bx					
a=81.84285714					
b=1.572857143					
r ² =.9506094419					
r=.9749920215					

□

Note

- (1) Notice that the value of r calculated in (i) and (ii) are the same. This illustrates an important property of r : it is **independent of the scale of measurement** for temperature.
- (2) Students could also use **4: LinReg(ax+b)** to calculate r from the GC.

6.6 # Linear Regression

In the last section, we used both the scatter diagram and the product moment correlation coefficient between the two variables to indicate whether it is meaningful to model the observed data with a straight line. If it appears that the data fits into a linear model, we then attempt to find an equation to represent the relationship by linear regression.

In Example 4, the speed, v , is controlled and the temperature, T , is measured based on v . Thus, v is known as the **independent variable** while T , whose value depends on v , is called the **dependent variable**. In this case, when we attempt to find an equation to represent the relationship between T and v , we should express T in terms of v .

On the other hand, in Example 3, we were investigating the number of sit ups and push-ups a student can do. In this case, there is no clear dependency between the two variables. Therefore, when we attempt to find an equation to represent the relationship between the two variables, we can express the number of sit ups in terms of numbers of push-ups, or the other way around.

6.7 Method of Least Squares

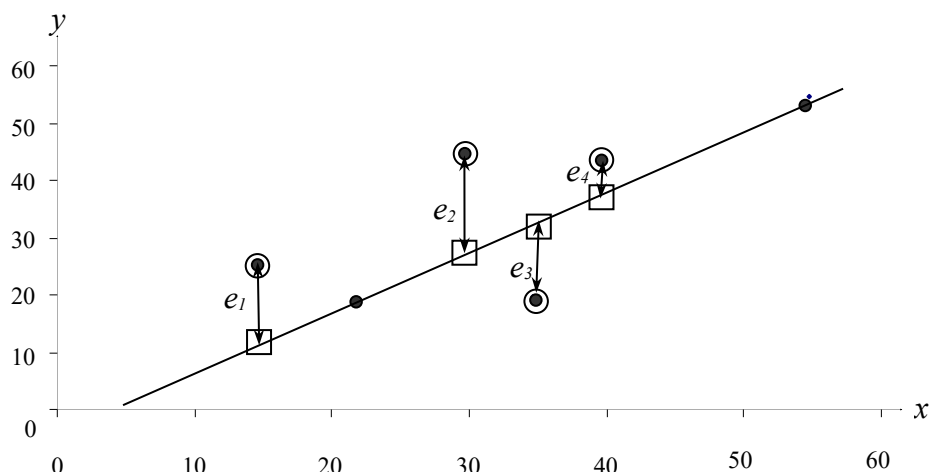
Consider the data given in the scatter diagram below. We randomly draw a line to fit the data first. The line drawn below may not be the line that best represents the data, i.e., the line of best fit. There are several ways to find a line of best fit and we use the method most commonly used for finding such a line called the **least squares method**. The line obtained by this method is called the **least squares regression line**.

We consider x as the independent variable and y as the dependent variable. The circled points in the diagram correspond to observed data points. If we were to use the line $y = a + bx$ to model the data, we would predict a different y -value for the corresponding x -value. The difference is the error, e , which is known as the **residual** is calculated as

$$e = \text{observed } y \text{ value} - \text{the predicted } y \text{ value.}$$

Each pair of observations (x_i, y_i) produces a residual, e_i , for $i = 1, 2, \dots, n$. $\sum_{i=1}^n e_i^2$ is known as the **sum of the squares of the residuals** and $\sum_{i=1}^n e_i^2$ is denoted as $\sum_{i=1}^n e_i^2$.

The line $y = a + bx$ is the least square regression line when **the sum of the squares of the residuals in the vertical direction is a minimum**.



The least squares regression line of **y on x** is the line that produces the smallest $\sum e_i^2$, where x is the independent variable and y is the dependent variable.

6.8 Equation of the Least Squares Regression Line of y on x

Consider a line of equation $y = a + bx$. Given a set of data, we want to find the values of a and b that minimize

$$\begin{aligned}\sum e_i^2 &= \sum (\text{observed } y \text{ value} - \text{predicted } y \text{ value})^2 \\ &= \sum (y_i - (a + bx_i))^2.\end{aligned}$$

It can be proven that in order to minimise $\sum e_i^2$ (see Appendix C),

$$b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} \quad \text{and} \quad a = \bar{y} - b\bar{x} \quad \text{----- (1)}$$

Consider the general equation of a line $y = a + bx$ and substituting (1) into the equation,

$$\begin{aligned}y &= (\bar{y} - b\bar{x}) + bx \\ \Rightarrow y - \bar{y} &= b(x - \bar{x})\end{aligned}$$

Therefore, the **equation of the regression line of y on x** is (given in MF27)

$$y - \bar{y} = b(x - \bar{x}) \quad , \quad \text{where} \quad b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

b is also known as the **regression coefficient of y on x**.

Note

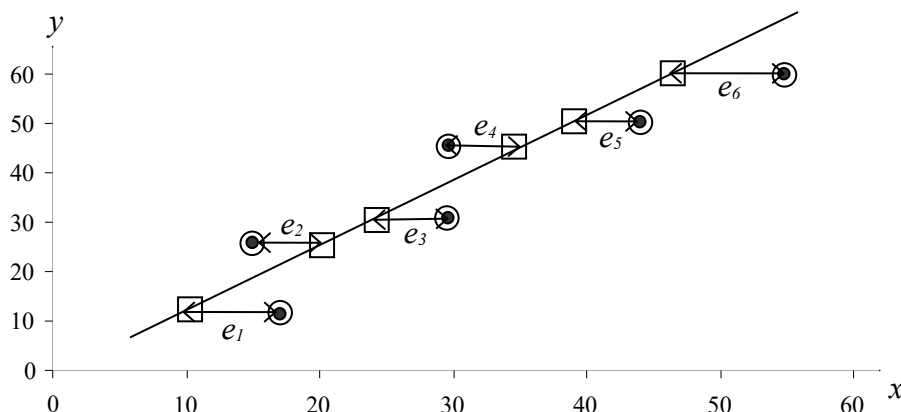
1. $\sum (y - \bar{y})(x - \bar{x}) = \sum xy - \frac{\sum x \sum y}{n}$.
2. $\sum (x - \bar{x})^2 = \sum x^2 - \frac{(\sum x)^2}{n}$.
3. Another formula for $b = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$.

6.9 Equation of the Least Squares Regression Line of x on y

Now we consider y as the independent variable and x as the dependent variable. We need to find the equation of the regression line of x on y , i.e., $x = c + dy$ where c and d are to be found. In this case, the residual, e , is calculated as

$$e = \text{observed } x \text{ value} - \text{the predicted } x \text{ value.}$$

The line $x = c + dy$ is the least square regression line when **the sum of the squares of the residuals in the horizontal direction is a minimum.**



We can show that the **equation of the regression line of x on y** is

$$x - \bar{x} = d(y - \bar{y}) \quad , \quad \text{where} \quad d = \frac{\sum (y - \bar{y})(x - \bar{x})}{\sum (y - \bar{y})^2}.$$

Note

1. $\sum (y - \bar{y})^2 = \sum y^2 - \frac{(\sum y)^2}{n}$.

2. Another formula for d is $d = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sum y^2 - \frac{(\sum y)^2}{n}}$.
3. The equation of the regression line of x on y **cannot be** found by making x the subject in the equation of the regression line of y on x . (and vice versa)

6.10 Use of Regression Lines for Estimation

Consider the data (x, y) . Given the value of one of the variables, regression lines can be used to predict or estimate the value of the other. The choice of the regression line used depends on the context of the situation:

- (a) If there is a clear indication that x is the independent variable, we will always use the regression line of y on x to do estimation.
- (b) For cases where there is no clear independent variable, if we want to estimate y for a given value of x , we use the regression line of y on x . If we want to estimate x for a given value of y , use the regression line of x on y .

Note

1. When we do estimation within the given range of values of the data, it is known as **interpolation**.
2. When we do estimation outside the given range of values of the data, it is known as **extrapolation**. Values obtained by extrapolation may not be reliable since the relationship between the two variables may not follow the same linear model outside the range of the data.
3. Estimates using regression lines are only reliable if both the following conditions are met:
 - (a) The value of r of the data is close to ± 1 (and the scatter diagram also suggests that there is a strong linear correlation).
 - (b) The estimation is done within the given range of values of data.

Example 5

In a physical education class, the number of push-ups (x) and sit-ups (y) done by a sample of ten randomly chosen students were recorded in the table below.

Student	1	2	3	4	5	6	7	8	9	10
x	27	22	15	35	30	52	35	55	40	40
y	30	26	25	42	38	40	32	54	50	43

- Find the equation of the regression line of y on x .
- Interpret the slope and intercept in the context of the question.
- Find the equation of the regression line of x on y .
- Use a suitable line to predict the number of sit-ups a student can do when he can do 50 push-ups. Give a reason if the predicted value is reliable.
- Give a reason whether it is reliable to use the equation in (i) to predict the number of sit-ups when 60 push-ups are done.

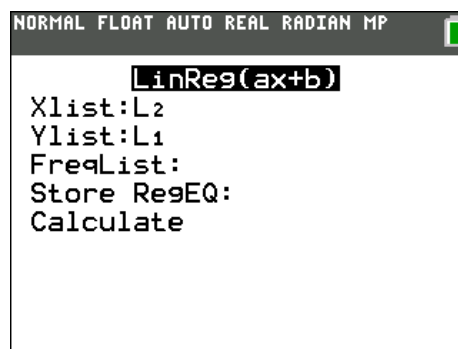
Solution

- From GC, the equation of the regression line of y on x is $y = 0.658x + 14.9$ (to 3 s.f.).
[Refer to GC screenshot in Example 3 solution]
- Slope: For every increase of 10 push-ups, it is estimated that there will be an expected increase of about 6.6 sit-ups.
 y -intercept $(0, 15)$: When a student cannot do any push-ups, the student can still do about 15 sit-ups.
- From GC, the equation of the regression line of x on y is $x = 1.07y - 5.60$ (to 3 s.f.).

- ① Store data under L_1 (x) and L_2 (y).

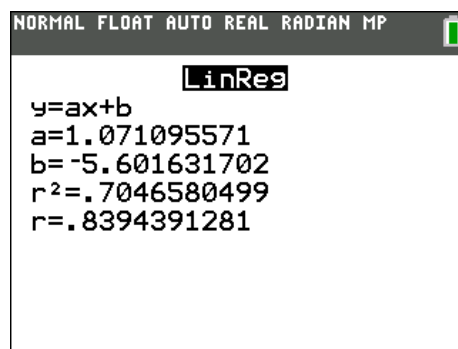
NORMAL FLOAT AUTO REAL RADIAN MP					
L1	L2	L3	L4	L5	2
27	30	-----	-----	-----	
22	26				
15	25				
35	42				
30	38				
52	40				
35	32				
55	54				
40	50				
40	43				
-----	-----				
L2(10) = 43					

- ② Press **[stat]** and select CALC. Select **4:LinReg(ax+b)** and press **[enter]**.
- ③ Key in L_2 under Xlist, then L_1 under Ylist and press Calculate.



From the results displayed,

the equation of the regression line of x on y is $x = 1.07y - 5.60$.



- (iv) Since there is no clear independent variable and we want to estimate y for a given value of x , we use the regression line of y on x .

When $x = 50$, $y = 0.657885(50) + 14.908225 = 47.8$ (to 3 s.f.).

He can do 47.8 sit ups ≈ 48 sit ups. Since $x = 50$ is within the data range of x and $r = 0.839$ (to 3 s.f.) is close to 1, then there is a strong positive linear correlation between x and y . Therefore the estimated value of y obtained should be reliable.

Since $x = 50$ is within the data range, we have to go on to comment on the strength of the correlation between x and y when determining the reliability.

- (v) No, because 60 is outside the data range $[15, 55]$ (**extrapolation**), hence it is not reliable.

Since $x = 60$ is outside the data range, it is not reliable to do extrapolation, hence there is no need to comment on the strength of the correlation between x and y .

Example 6

An electrical fire was switched on in a cold room and the temperature of the room was noted at 5-minute intervals.

Time, x (in minutes) from switching on fire	0	5	10	15	20	25	30	35	40
Temperature y , c (in $^{\circ}\text{C}$)	0.4	1.5	3.4	5.5	7.7	9.7	11.7	13.5	15.4

- Find the equation of the regression line of y on x and the product moment correlation coefficient between x and y . Comment on what its value implies about the relationship between x and y .
- Explain why the regression line of y on x rather than the regression line of x on y should be used to predict the time that has passed after switching on the fire if the temperature is 9°C .
- Predict the temperature of the room when the fire is switched on for 28 and 60 minutes. Comment on the reliability of your answers.
- By using the equation of the regression line of y on x , deduce the equation of the regression line of
 - y on t where y is the temperature in $^{\circ}\text{C}$ and t is time in hours,
 - z on x where z is the temperature in Kelvin (K) and x is time in minutes.

(A temperature in $^{\circ}\text{C}$ is converted to K by adding 273)
- Comment on the values of r obtained in (d)(i) and (d)(ii).

Solution

- Using GC, the required equation of the regression line is $y = 0.389x - 0.142$ (to 3 s.f.). The product moment correlation coefficient, r , between x and y is 0.999 (to 3 s.f.). Since r is very close to 1, this indicates a strong positive linear correlation between x and y .
- From the question, x is the controlled variable and y is measured based on regular time interval of 5 minutes, suggesting that y is dependent on x . Thus, the regression line of y on x should be used.
- Using $y = 0.38933x - 0.14222$, when $x = 28$, $y = 10.8^{\circ}\text{C}$, and when $x = 60$, $y = 23.2^{\circ}\text{C}$. As $r = 0.999$ is very close to 1 and 28 is within the data range of x , the estimated temperature of 10.8°C when $x = 28$ can be said to be reliable. However, for the second case, when $x = 60$, this value is outside the data range of x , which means the estimated temperature of 23.2°C obtained from the equation of the regression equation may be unreliable.

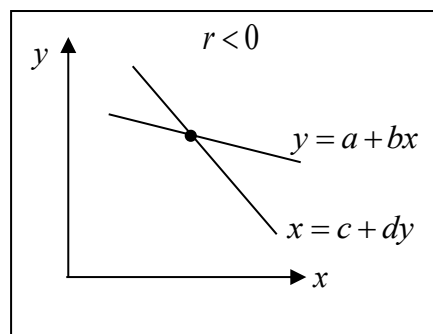
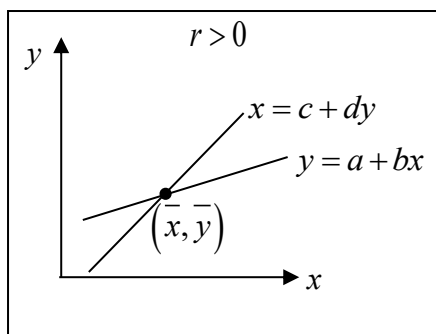
- (d) (i) Since $t = \frac{x}{60} \Rightarrow x = 60t$.
Substitute this result into $y = 0.38933x - 0.14222$ and we obtain
 $y = 23.4t - 0.142$.
- (ii) Using the relation $z - y + 273 \Rightarrow y = z - 273$.
Substitute this result into $y = 0.38933x - 0.14222$ and we have
 $z = 0.389x + 273$.
- (e) $r = 0.999$ for both (i) and (ii) as r is independent of the scale of measurement of x and y . $\square$

6.11 # Properties of Regression Lines

In general, the regression line of y on x , i.e., $y = a + bx$, is different from the regression line of x on y , i.e., $x = c + dy$.

These are some observations about the lines and r :

- (i) Both the regression line of y on x as well as the regression line of x on y pass through the sample mean $(\bar{x}, \bar{y})$.
- (ii) If $r > 0$, both regression coefficients b and d are positive.
If $r < 0$, both regression coefficients b and d are negative.



- (iii) The larger the value of $|r|$, i.e., almost 1, the “closer” the regression line of y on x is to the regression line of x on y .
If $r = \pm 1$, the regression lines of y on x and x on y are identical. All data points lie on the line.

6.12 Transformations

Not all correlation between x and y are linear. If the correlation between x and y is not linear, we can sometimes use a suitable transformation to linearise the correlation. Here are some examples:

Relationship	Transformation	Linear Relationship
$y = ax^b$	Take natural logarithm (or take logarithm of another base)	$\ln y = \ln a + b \ln x$ i.e., $\ln y$ and $\ln x$ have a linear relationship.
$y = ae^{bx}$	Take natural logarithm	$\ln y = \ln a + bx$ i.e. $\ln y$ and x have a linear relationship.
$y = \sqrt{ax + b}$	Square both sides	$y^2 = ax + b$ i.e. y^2 and x have a linear relationship.
$y = \frac{1}{ax + b}$	Take reciprocal	$\frac{1}{y} = ax + b$ i.e., $\frac{1}{y}$ and x have a linear relationship.

Example 7

The following data were collected during an experiment which investigated the average lifespan of plants, t days, as the pH, y , of the soil in which the plant was grown varied.

y	4.5	5.2	6.1	6.5	7.0	7.3	8.5	9.5
t	1.14	1.20	1.26	1.29	1.33	1.35	1.42	1.48

It is thought that the lifespan of the plant and the pH of the soil can be modelled by one of the formulae

$$t = a + by, \quad \text{or} \quad \ln t = \ln a + b \ln y$$

where a , b , c and d are constants.

- Find, correct to 4 decimal places, the value of the product moment correlation coefficient between
 - t and y ,
 - $\ln t$ and $\ln y$.
- Explain which of $t = a + by$ and $\ln t = \ln a + b \ln y$ is the better model and hence find the values of a and b for this model.

Hence $\ln a = -0.3952 \Rightarrow a = 0.674$ (to 3 s.f.) and $b = 0.349$ (to 3 s.f.).

- NORMAL FLOAT AUTO REAL RADIAN MP
LinReg
 $y = ax + b$
 $a = .3489057024$
 $b = -.3952019647$
 $r^2 = .9990631963$
 $r = .9995314884$

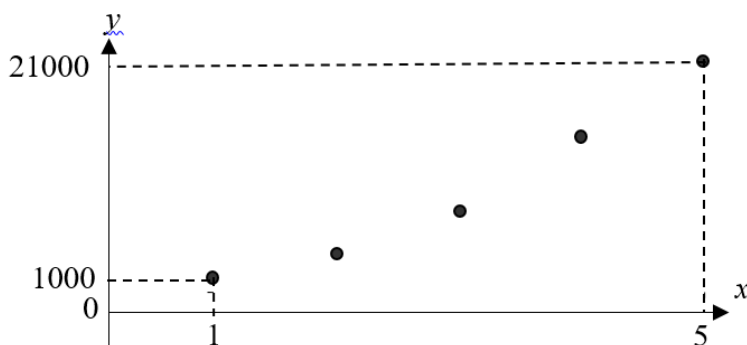
N-20

Example 8

A bookshop sells a popular guidebook. The sales of the guidebook in each of five successive years (Yr 1 – 5) are given in this table.

Year (x)	1	2	3	4	5
Sales (y)	1000	3000	7000	14000	21000

The value of the product moment correlation coefficient is 0.976 correct to 3 significant figures. The scatter diagram for the data is shown below.



- (a) The bookshop owner attempts to model the relationship between y and x with a straight line. Explain whether this is unlikely to provide a good model.

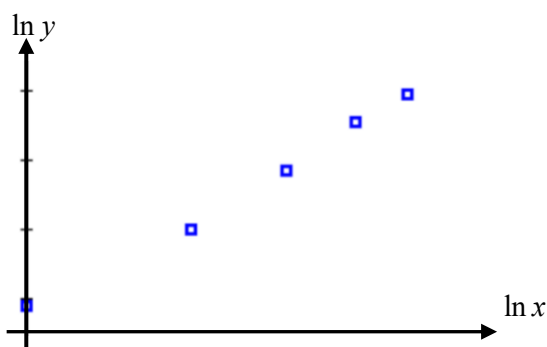
The owner now tries a model in which y has been transformed to $\ln y$, and x has been transformed to $\ln x$.

- (b) (i) Sketch a scatter diagram of $\ln y$ against $\ln x$ for the data given in the table.
 (ii) The owner models the data with the equation $\ln y = a + b \ln x$. Find the values of the constants a and b and state the value of the product moment correlation coefficient for this model.
- (c) Comment on the owner's two models.

Solution

- (a) The scatter diagram shows that as x increases, y increases at an increasing rate (curvilinear relationship, curve concaves upwards). Therefore a linear model is not the most suitable one.

- (b) (i)



- ① Key in x and y data into L_1 and L_2 respectively. Scroll up to the header L_3 and highlight it by pressing enter.

② Key in the transformations, in this case $L_3 = \ln(L_1)$ and $L_4 = \ln(L_2)$.
Press to generate the transformed data.

③ Find the equation of the regression line using L_3 and L_4 .

NORMAL FLOAT AUTO REAL RADIAN MP
LinReg
 $y=a+bx$
 $a=6.815631664$
 $b=1.919318251$
 $r^2=.9932916127$
 $r=.9966401621$

(c) From the scatter diagram in the question, the points appear to lie closely to a curve than a straight line. Furthermore, the product moment correlation coefficient for the linear model is 0.976 (3 s.f.) whereas the product moment correlation coefficient for the second model is 0.997 (3 s.f.), which is closer to 1.

Hence the second model with the equation $\ln y = a + b \ln x$ is more suitable.

The value of r given may be very close to 1 but it does not necessarily imply that a linear model $y = a + bx$ is the most suitable for the data. It is always important to decide which model is more suitable based on the scatter diagram.

Summary

- A scatter diagram is obtained when each of the observed bivariate data (x_i, y_i) , $i = 1, 2, \dots, n$ is plotted on the Cartesian plane.
- The product moment correlation coefficient (in MF27) is defined as:

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}} = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sqrt{\left(\sum x^2 - \frac{(\sum x)^2}{n} \right) \left(\sum y^2 - \frac{(\sum y)^2}{n} \right)}}$$

- In order to determine if there is a linear relationship, a scatter diagram, together with the product moment correlation coefficient should be used.
- $-1 \leq r \leq 1$
- When $r > 0$, there is positive linear correlation between x and y .
When $r < 0$, there is negative linear correlation between x and y .
When $r = 0$, there is no linear correlation between x and y .
- r is independent of the scale of measurement of x and y .
- The least squares regression line of y on x is the line that produces the smallest $\sum e_i^2$ in the vertical direction.
- The least squares regression line of x on y is the line that produces the smallest $\sum e_i^2$ in the horizontal direction.
- Equation of the regression line of y on x is

$$y - \bar{y} = b(x - \bar{x}) \quad , \quad \text{where } b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}} .$$

- Equation of the regression line of x on y is

$$x - \bar{x} = d(y - \bar{y}) \quad , \quad \text{where } d = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (y - \bar{y})^2} = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sum y^2 - \frac{(\sum y)^2}{n}} .$$

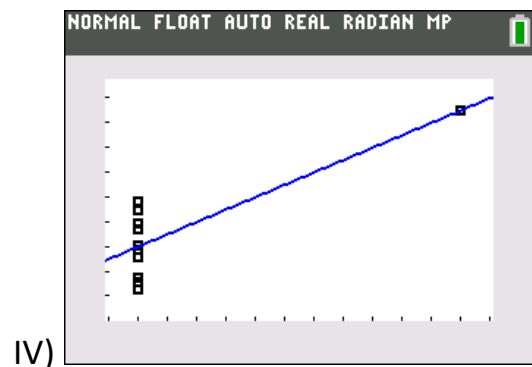
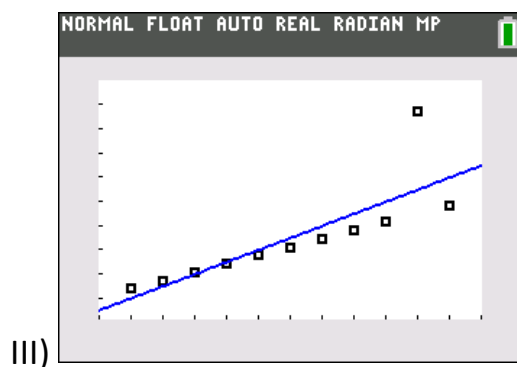
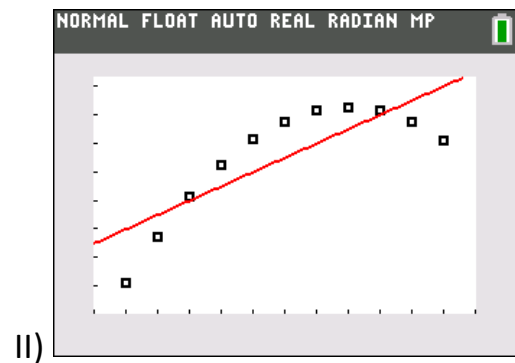
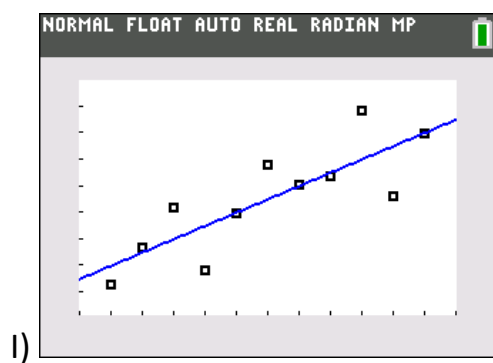
- If there is a clear independent variable x , we will always use the regression line of y on x to do estimation.
- For cases where there is no clear independent variable, if we want to estimate y for a given value of x , we use regression line of y on x . If we want to estimate x for a given value of y , use regression line of x on y .
- Both the regression line of y on x as well as the regression line of x on y pass through the sample mean $(\bar{x}, \bar{y})$.

Appendix A – Anscombe Quartet

To illustrate the importance of using the scatter diagram to support the product moment correlation coefficient, F.J Anscombe constructed the following 4 data sets which have exactly the same r value but appear very different when graphed:

Anscombe's Quartet							
I		II		III		IV	
x	y	x	y	x	y	x	y
10.0	8.04	10.0	9.14	10.0	7.46	8.0	6.58
8.0	6.95	8.0	8.14	8.0	6.77	8.0	5.76
13.0	7.58	13.0	8.74	13.0	12.74	8.0	7.71
9.0	8.81	9.0	8.77	9.0	7.11	8.0	8.84
11.0	8.33	11.0	9.26	11.0	7.81	8.0	8.47
14.0	9.96	14.0	8.10	14.0	8.84	8.0	7.04
6.0	7.24	6.0	6.13	6.0	6.08	8.0	5.25
4.0	4.26	4.0	3.10	4.0	5.39	19.0	12.50
12.0	10.84	12.0	9.13	12.0	8.15	8.0	5.56
7.0	4.82	7.0	7.26	7.0	6.42	8.0	7.91
5.0	5.68	5.0	4.74	5.0	5.73	8.0	6.89

Note that all 4 sets of data have the same mean, variance, r value and linear regression line.



Appendix B – Properties of r

The measure, r will not change if we add a constant or multiply a positive constant to all the values of a variable.

if $s = a + bx$, $t = c + dy$, where a, b, c, d are constants, b and d both positive, then

$$r_{st} = r_{xy}.$$

$$\begin{aligned} r_{st} &= \frac{\sum (s - \bar{s})(t - \bar{t})}{\sqrt{\sum (s - \bar{s})^2 \sum (t - \bar{t})^2}} \\ &= \frac{\sum [(a + bx) - (a + b\bar{x})][(c + dy) - (c + d\bar{y})]}{\sqrt{\sum [(a + bx) - (a + b\bar{x})]^2 \sum [(c + dy) - (c + d\bar{y})]^2}} \\ &= \frac{\sum (bx - b\bar{x})(dy - d\bar{y})}{\sqrt{\sum (bx - b\bar{x})^2 \sum (dy - d\bar{y})^2}} = \frac{bd \sum (x - \bar{x})(y - \bar{y})}{\sqrt{b^2 d^2 \sum (x - \bar{x})^2 \sum (y - \bar{y})^2}} \\ &= \frac{bd \sum (x - \bar{x})(y - \bar{y})}{bd \sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}} \quad (\text{Note } \sqrt{b^2 d^2} = bd \because bd \text{ is positive.}) \\ &= \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}} = r_{xy} \end{aligned}$$

Appendix C – Partial Differentiation to Determine Regression Coefficients

Let $y_1, y_2, \dots, y_n$ be the observed values corresponding to the values $x_1, x_2, \dots, x_n$.

The least squares line $y = a + bx$ is the line that minimises the sum of the squares of

the residuals ie. $S = \sum e_i^2 = \sum_{i=1}^n (y_i - a - bx_i)^2$.

Calculating the partial derivative of S with respect to a :

$$\begin{aligned} \frac{\partial S}{\partial a} &= -2 \sum (y_i - a - bx_i) \quad (\text{note the notation } \frac{\partial S}{\partial a} \text{ is being used instead of } \frac{dS}{da}) \\ &= -2 \left(\sum y_i - na - b \sum x_i \right) \end{aligned}$$

$$\text{Let } \frac{\partial S}{\partial a} = 0 \text{ to get } a = \frac{\sum y_i}{n} - b \frac{\sum x_i}{n} = \bar{y} - b\bar{x}.$$

Calculating the partial derivative of S with respect to b :

$$\frac{\partial S}{\partial b} = -2 \sum x_i (y_i - a - bx_i) \text{ and let } \frac{\partial S}{\partial b} = 0.$$

$$\text{Thus } \sum x_i y_i - a \sum x_i - b \sum x_i^2 = 0 \Rightarrow \sum x_i y_i - (\bar{y} - b\bar{x})n\bar{x} - b \sum x_i^2 = 0.$$

$$\Rightarrow b = \frac{\sum x_i y_i - n\bar{x}\bar{y}}{\sum x_i^2 - n\bar{x}^2} \Rightarrow b = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}.$$



Statistics Tutorial 6

1. The product moment correlation coefficient is denoted by r . Comment on the validity of following statements:
 - (a) $r = 0$ for a set of data (x, y) implies that X and Y are unrelated.
 - (b) If X is the number of cigarettes smoked per day by people dying of lung cancer and Y is the age at death, then $r = -0.9$ implies that the smoking more cigarettes per day causes a person to die younger.
 - (c) The value of r for a set of sample data (x, y) is 1 means that a linear relation holds for the population (X, Y) .

2.
 - (a) The product moment correlation coefficient for a set of data (x, y) is denoted by r .
 - (i) State a data point which can be included such that r will remain the same for the new set of data.
 - (b) Express, in terms of r ,
 - (i) the product moment correlation coefficient of (y, x) .
 - (ii) the product moment correlation coefficient when all values of x are increased by 5.
 - (iii) the product moment correlation coefficient for the set of $(x, -y)$.
 - (c) **[A-Level N2017/II/8(a)]** Draw separate scatter diagrams, each with 8 points, all in the first quadrant, which represent the situation where the product moment correlation coefficient between the variables x and y is
 - (i) -1 ,
 - (ii) 0 ,
 - (iii) Between 0.5 and 0.9 .[3]
 - (d) **[A-Level N2014/II/8(a)]** Sketch a scatter diagram that might be expected when x and y are related approximately by $y = px^2 + t$ in each of the cases (i) and (ii) below. In each case your diagram should include 6 points, approximately equally spaced with respect to x , and with all x -values positive.
 - (i) p and t are both positive
 - (ii) p is negative and t is positive.[2]

3. With the aid of a suitable diagram, describe the difference between the regression line of Y on X and that of X on Y . [3]

Under what circumstances would the above two lines be co-incident? [1]

The following summarizes the data from 10 sets of lengths(x) and breadths(y) in mm:

$$\sum x = 1782, \sum y = 1483, \sum x^2 = 318086, \sum y^2 = 220257, \sum xy = 264582$$

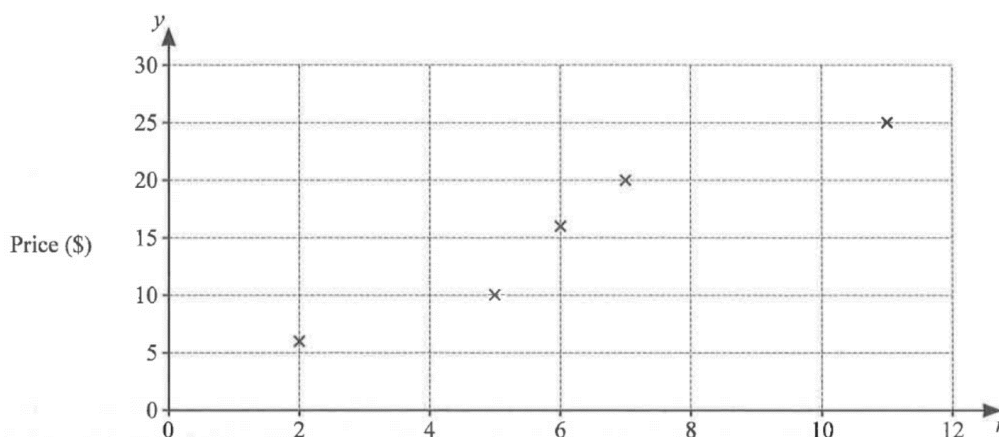
- Find the value of the product moment correlation coefficient. [1]
- Find the equation of the regression line of y on x and that of x on y . [2]
- Predict the breadth when the length is 185 mm. Comment on the reliability of your prediction. [2]

4. [A Level 2024/P2/Q11(b)]

- (b) Kai is investigating the relationship between the number, n , of a particular type of high-performance batteries in a pack and the price, \$ y , of the pack she found in different stores. Her results are shown in the table below.

n	2	5	6	7	11
y	6	10	16	20	25

This information is shown in the scatter diagram below.



Kai decides to investigate whether the model $y = \frac{5}{2}n + 1$ is a good fit for this data.

- Draw the line $y = \frac{5}{2}n + 1$ on the scatter diagram above. [1]

- (ii) For the model $y = f(n)$, the residual for a point (a, b) is $b - f(a)$.
- (A) Mark the residuals for the points on the scatter diagram above. [1]
- (B) Explain why Kai should use the sum of the squares of the residuals rather than the sum of the residuals when assessing the fit of the model. [1]
- (C) Calculate the sum of the squares of the residuals for the line $y = \frac{5}{2}n + 1$. [1]

Kai's friend points out that other lines parallel to Kai's line can be drawn which are a better fit for the data.

- (iii) Explain how the sum of the squares of the residuals for a line that is a better fit for the data differs from the sum of the squares of the residuals for the line $y = \frac{5}{2}n + 1$ found in part (ii)(C). [1]
- (iv) Find the range of values of c for which the line $y = \frac{5}{2}n + c$ is a better fit for the data than the $y = \frac{5}{2}n + 1$. [3]

5. **JPJC Prelim 9758/2023/02/Q9**

The shear strength of soil is the ability of the soil to resist tensional stresses and sustain loads. The water content of soil is the ratio of the weight of water to the weight of the solids in a mass of soil. A soil investigation laboratory collected soil from a reclaimed site to study the effect of water content on shear strength. The information gathered about the shear strength P , measured in KiloPascal (KPa), at different water content w , measured in percentage is presented in the table.

Water Content, w (%)	10.0	12.5	15.0	17.5	20.0	22.5	25.0	27.5
Shear Strength, P (KPa)	2500	2100	1820	1630	1500	1410	1320	1270

- (i) Draw a scatter diagram of these data. Use your diagram to explain whether the relationship between P and w is likely to be well modelled by an equation of the form $P = aw + b$, where a and b are constants. [2]
- (ii) By calculating the relevant product moment correlation coefficients, determine whether the relationship between P and w is modelled better by $P = aw + b$ or $P = a\sqrt{w} + b$. Explain how you decide which model is better, and state the equation in this case. [5]

- (iii) Use your equation to estimate the shear strength when the water content is 40%. Explain whether your estimate is reliable. [2]
- (iv) Subsequently, it is known that one data point is not presented in the information table and that the inclusion of this data point does not affect the appropriate equation in (ii). State the missing data point, giving your answer correct to the nearest integer. [1]
- (v) Given that 1 MegaPascal = 1000 KiloPascal, re-write your equation in (ii) so that it can be used when the shear strength is measured in MegaPascal. Explain whether the corresponding product moment correlation coefficient would differ from that calculated (ii). [2]

6. DHS Prelim 9758/2023/02/Q11

- (a) Sketch a scatter diagram that might be expected when x and y are related approximately as given in each of the cases (A) and (B) below. In each case, your diagram should include 6 points, approximately equally spaced with respect to x , and with all x - and y -values positive. The letters a , b , c and d represent constants.

(A) $y = a + b\sqrt{x}$, where a is positive and b is negative,

(B) $y = c + \frac{d}{x}$, where c is positive and d is negative. [2]

- (b) The data below shows the diameter h , in cm, of coconuts of 9 different ages, in k months.

Age in months (k)	1	2	3.5	4.5	5	7.5	9	10.5	12
Diameter in cm (h)	6	9.5	13	15	16.5	17.5	18.5	19	19.5

- (i) Calculate the value of the product moment correlation coefficient r between k and h . State, with a reason, whether the value of r would be different if the diameter is measured in inches instead. [2]
- (ii) Give a sketch of the scatter diagram for the data. [1]
- (iii) It is desired to predict the diameter of coconuts in the long run. Explain why neither a linear nor a quadratic model is likely to be appropriate. [2]

It is suggested that diameter h can be modelled by the formula $h = m + n \ln k$.

- (iv) Find the equation of the regression line for the suggested model, and the value of the product moment correlation coefficient between $\ln k$ and h . [2]
- (v) Estimate, to 1 decimal place, the age of a coconut with a diameter of 13.7 cm, and comment on the reliability of your answer. [2]
- (vi) Given that 1 inch = 2.54 cm, state the equation of regression line if the diameter is measured in inches instead of cm. [1]

7. VJC Prelim 9758/2022/02/Q8

An agricultural experiment was carried out to study the effect of a certain fertilizer on the growth of seedlings. The fertilizer is applied at various concentrations to a random sample of ten plots of land. Seeds are sown and two weeks later, the mean height of the seedlings on each plot of land is measured. The results are shown in the table.

Concentration of fertilizer (x grams/m ²)	5	10	20	30	40	50	60	70	80	90
Mean height of seedlings (y cm)	4.2	9.0	15.6	18.5	19.2	22.5	24.0	25.4	25.4	26.2

- (i) Draw the scatter diagram for these values, labelling the axes clearly. [1]

It is thought that the mean height of seedlings y can be modelled by one of the formulae

$$y = a + bx \quad \text{or} \quad y = c + d \ln x,$$

where a, b, c and d are constants.

- (ii) Find, correct to 4 decimal places, the value of the product moment correlation coefficient between

(a) x and y ,

(b) $\ln x$ and y . [2]

- (iii) Use your answers to parts (i) and (ii) to explain which of $y = a + bx$ or $y = c + d \ln x$ is the better model. [1]

It is required to estimate the value of x for which $y = 20.0$.

- (iv) Explain why neither the regression line of x on y nor the regression line of $\ln x$ on y should be used. [1]

- (v) Find the equation of a suitable regression line and use it to find the required estimate. [2]

- (vi) Re-write your equation from part (v) so that it can be used when y , the mean height of seedlings, is given in mm. [1]

8. TMJC JC2 Prelim 9758/2019/02/Q8

In a chemical reaction, the amount of catalyst used, x grams, and the resulting reaction times, y seconds, were recorded and the results are given in the table.

x	2.0	2.5	3.0	3.5	4.0	4.5	5.0	5.5	6.0
y	62.1	51.2	44.1	39.1	35.0	k	33.0	31.4	29.5

The equation of the regression line of y on x is $y = 68.8067 - 7.12667x$, correct to 6 significant figures.

(i) Show that $k = 37.3$, correct to 1 decimal place. [2]

(ii) Draw a scatter diagram for these values, labelling the axes clearly. [1]

It is suggested that the relationship between x and y can be modelled by one of the following formulae

(A) $y = a + bx$

(B) $y = c + dx^2$

(C) $y = e + \frac{f}{x}$

where a , b , c , d , e and f are constants.

(iii) Find the value of the product moment correlation coefficient for each model. Explain which is the best model and find the equation of a suitable regression line for this model. [5]

(iv) By using the equation of the regression line found in part (iii), estimate the reaction time when the amount of catalyst used is 4.2 grams. Comment on the reliability of your estimate. [2]

9.

[A-Level N2019/II/10 (modified)]

Abi and Bhani find the fuel consumption for a car driven at different constant speeds. The table shows the fuel consumption, y kilometres per litre, for different constant speeds, x kilometres per hour.

x	40	45	50	55	60
y	22	20	18	17	16

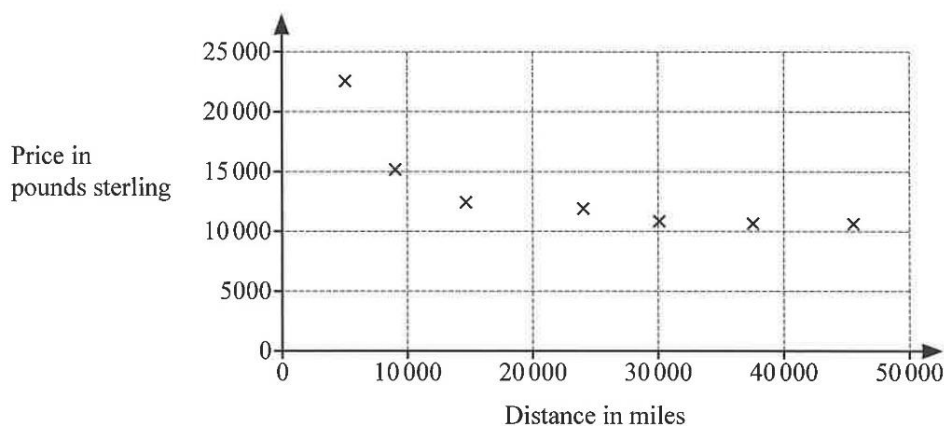
- (i) Abi decides to model the data using the line $y = 35 - \frac{1}{3}x$.
- (a) Using a piece of graph paper
- draw a scatter diagram of the data,
 - draw the line $y = 35 - \frac{1}{3}x$. [2]
- (b) For a line of best fit $y = f(x)$, the residual for a point (a, b) plotted on the scatter diagram is the vertical distance between $(a, f(a))$ and (a, b) . Mark the residual for each point on your diagram. [1]
- (c) Calculate the sum of the squares of the residuals for Abi's line. [1]
- (d) Explain why, in general, the sum of the squares of the residuals rather than the sum of the residuals is used. [1]
- Bhani models the same data using a straight line passing through the points $(40, 22)$ and $(55, 17)$. The sum of the squares of the residuals for Bhani's line is 1.
- (ii) State, with a reason, which of the two models, Abi's and Bhani's, gives a better fit. [1]
- (iii) State the coordinates of the point that the least squares regression line must pass through. [1]
- (iv) Use your calculator to find the equation of the least squares regression line of y on x . State the value of the product moment correlation coefficient. [3]
- (v) Use the equation of the regression line to estimate the fuel consumption when the speed is 30 kilometres per hour. Explain whether you would expect this value to be reliable. [2]
- Cerie performs a similar experiment on a different car. She finds that the sum of the squares of the residuals for her line is 0. What can you deduce about the data points in Cerie's experiment? [1]

10. [A-Level N2022/II/10]

Adriana is investigating how the prices of used cars depend on the distances the cars have been driven. She collects some information from a UK website about 7 cars, all of the same make and model. This is given in the table below; the distance, d , is in miles and the price, p , is in pounds sterling.

d	14588	8954	24 030	37 551	45 452	30 100	5055
p	12 450	15 200	11 900	10 700	10 600	10 900	22 500

A scatter diagram for the data is shown below.



The product moment correlation coefficient between d and p is -0.78 .

- (a) What conclusion should Adriana reach about the relationship between d and p ?
Justify your answer. [2]
- (b) Explain whether the product moment correlation coefficient would differ if Adriana converted distances to kilometres and the prices to Singapore dollars.
[1 mile = 1.6 kilometres; 1 pound sterling = 1.7 Singapore dollars] [1]

Adriana checks the original information. She realises that the car with $d = 5055$ and $p = 22\,500$ was a special version of the model, with a bigger engine and many extra features.

- (c) Explain why the data relating to this special car should not be used in the analysis of the data. [1]

In the rest of this question you should not include the data for this special car.

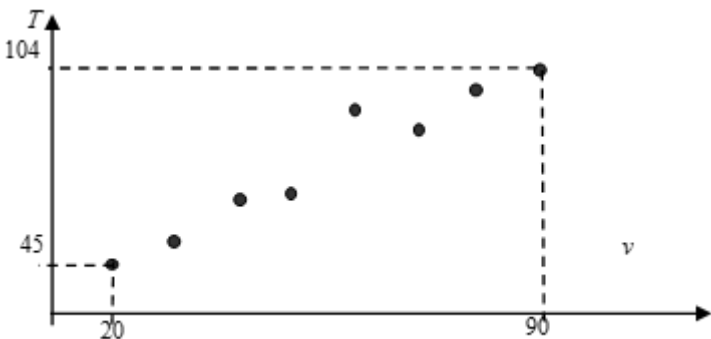
- (d) Using a reciprocal transformation, sketch a scatter diagram for p against $\frac{100\,000}{d}$. Draw, by eye, a line of best fit on your scatter diagram. [2]
- (e) Show on the scatter diagram in part (d) the distances which are used in drawing the least squares regression line of p on $\frac{100\,000}{d}$. Explain why these distances are squared, and why this is referred to as the 'method of least squares'. [3]
- (f) Use your calculator to find the equation of the least squares regression line of p on $\frac{100\,000}{d}$ and the value of the corresponding product moment correlation coefficient. [3]
- (g) Use your equation to estimate the price of a car of this make and model that has travelled 5055 miles. Explain whether you would expect this estimate to be reliable. [2]

Answers:

3. (i) 0.744 (ii) $y = 0.584x + 44.3$; $x = 0.949y + 37.4$ (iii) 152 mm
4. (b) (ii)(C) 26.75 (iv) $-1.2 < c < 1$
5. (ii) $P = -570\sqrt{w} + 4130$ (iii) 527 (iv) $w=19, P=1694$ (v) $P = -0.570\sqrt{w} + 4.13$
6. (b)(i) 0.911 (b)(iv) $h = 6.15 + 5.61 \ln k$, $r = 0.992$ (b)(v) 3.8 (b)(vi) $h = 2.42 + 2.21 \ln k$
7. (ii) (a) $r = 0.925588 \approx 0.9256$ (b) $r = 0.996583 \approx 0.9966$ (v) $y = -8.45 + 7.81 \ln x$, 38
(vi) $y = -84.5 + 78.1 \ln x$
8. (iii) $r = -0.922$, $r = -0.866$, $r = 0.990$, $y = 13.7 + \frac{94.3}{x}$ (iv) 36.1
9. (i)(c) $4/3$ (iii) (50, 18.6)
(iv) $y = -0.3x + 33.6$, -0.985 (v) 24.6 km/l, Not reliable
10. (f) $P = \frac{5.06 \times 10^7}{d} + 9400$ (g) 19400

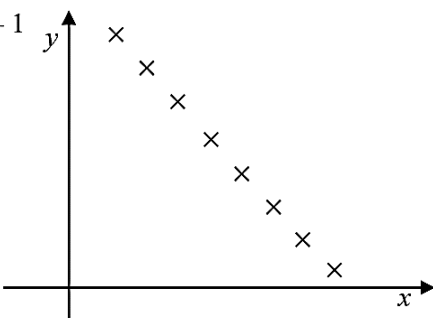
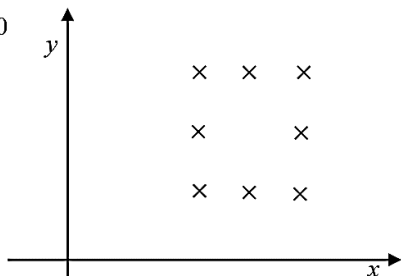
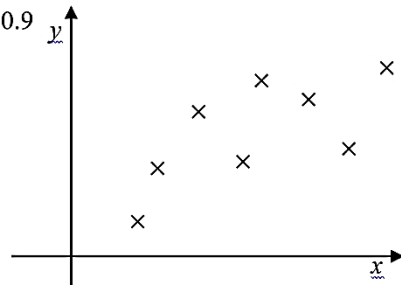
Skill Set

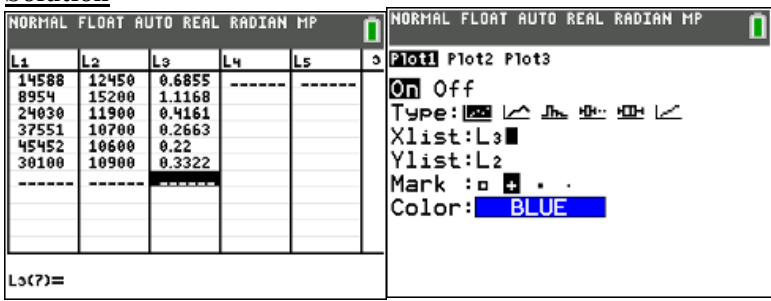
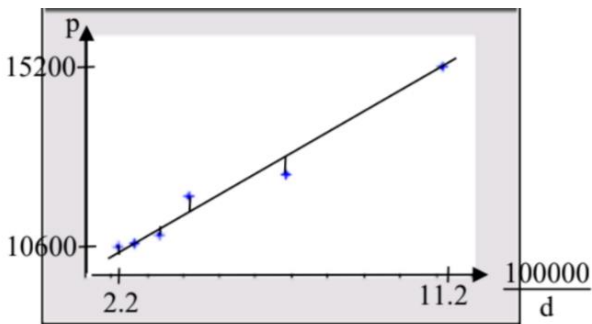
Chapter S6 Correlation and Regression

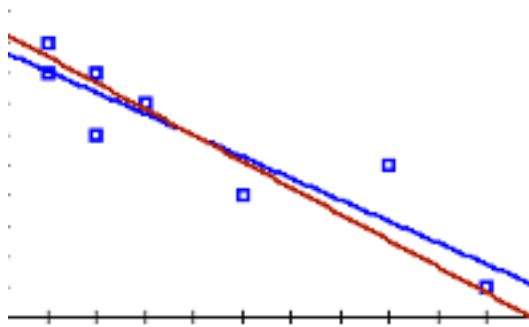
No.	Skills	Examples of questions involving the skills																		
1	Draw and use scatter diagram to determine if there is a plausible linear relationship between 2 bivariate variables	The temperature, T, in degree Celsius ($^{\circ}\text{C}$) of the tyre of a car is measured when the car travels at different speed, v (kmh^{-1}). Eight sets of data are obtained. Sketch the scatter diagram for the data. <table><tr><td>v</td><td>20</td><td>30</td><td>40</td><td>50</td><td>60</td><td>70</td><td>80</td><td>90</td></tr><tr><td>T</td><td>45</td><td>52</td><td>64</td><td>66</td><td>91</td><td>86</td><td>98</td><td>104</td></tr></table> Solution - In this case the speed is controlled and the temperature, T, is measured based on speed, v, of the car. Thus v is the independent variable while T, whose value depends on v is called the dependent variable. - Sketch such that the horizontal axis is for the independent variable and the vertical axis is for the dependent variable  The scatter diagram shows that there is a positive linear relationship between T and v.	v	20	30	40	50	60	70	80	90	T	45	52	64	66	91	86	98	104
v	20	30	40	50	60	70	80	90												
T	45	52	64	66	91	86	98	104												

No.	Skills	Examples of questions involving the skills																																				
2	Calculate and interpret the product moment correlation coefficient and use it as a measure of fit of a linear model together with the scatter diagram	- Use a GC to calculate and interpret the product moment correlation coefficient and use it as a measure of fit of linear model NORMAL FLOAT DEC RE <table><thead><tr><th>L1</th><th>L2</th><th>L3</th></tr></thead><tbody><tr><td>20</td><td>45</td><td></td></tr><tr><td>30</td><td>52</td><td></td></tr><tr><td>40</td><td>64</td><td></td></tr><tr><td>50</td><td>66</td><td></td></tr><tr><td>60</td><td>91</td><td></td></tr><tr><td>70</td><td>86</td><td></td></tr><tr><td>80</td><td>98</td><td></td></tr><tr><td>90</td><td>104</td><td></td></tr><tr><td>---</td><td>---</td><td></td></tr></tbody></table> L2(9)= NORMAL FLOAT DEC REAL RADIAN LinReg y=ax+b a=0.8738095238 b=27.69047619 r²=0.9506094419 r=0.9749920215 Since $r = 0.975$ (3 s.f.) is close to 1, there is a strong positive linear correlation between the temperature, T, and the speed, v, of the car. Therefore a linear model is suitable. - A scatter diagram together with the product moment correlation coefficient should be used to determine if there is a linear relationship. - In general,<table><tr><td>$0.8 < r < 1$</td><td>strong linear correlation</td></tr><tr><td>$0.5 < r < 0.8$</td><td>moderately strong linear correlation</td></tr><tr><td>$0 < r < 0.5$</td><td>weak linear correlation</td></tr></table> The above classification should only be treated as a guide. A value of r close to 1 does not necessarily imply one variable directly causes the other. - Use formula given in MF 27 to help calculate the product moment correlation coefficient when information on bivariate data is in summarized form. The following summarizes the data from 10 sets of lengths(x) and breadths(y) in mm: $\sum x = 1782, \sum y = 1483, \sum x^2 = 318086,$	L1	L2	L3	20	45		30	52		40	64		50	66		60	91		70	86		80	98		90	104		---	---		$0.8 < r < 1$	strong linear correlation	$0.5 < r < 0.8$	moderately strong linear correlation	$0 < r < 0.5$	weak linear correlation
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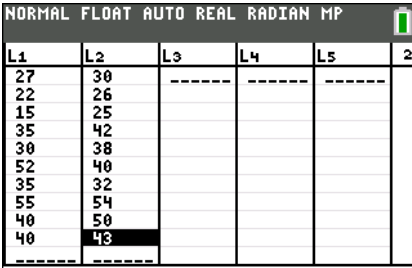
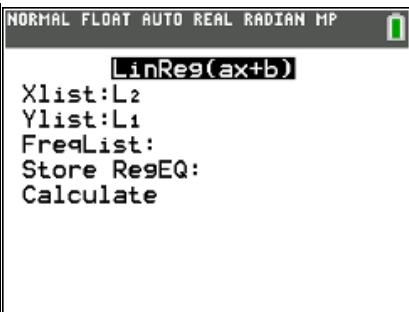
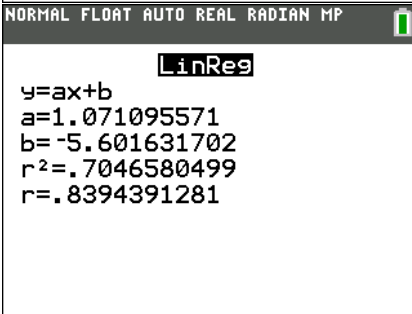
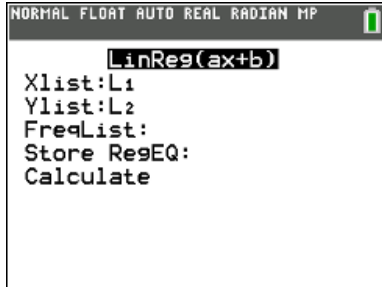
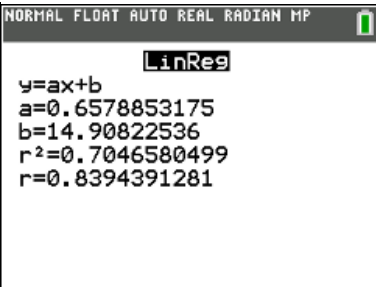
No.	Skills	Examples of questions involving the skills
		$\sum y^2 = 220257, \sum xy = 264582$ Find the value of the product moment correlation coefficient. <u>Solution</u> $r = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sqrt{\left(\sum x^2 - \frac{(\sum x)^2}{n}\right)\left(\sum y^2 - \frac{(\sum y)^2}{n}\right)}} = 0.744 \text{ (3s.f.)}$ • The product moment correlation coefficient is independent of the scale of measurement of the variables. This means that r will not change if we add a constant or multiply a positive constant to all the values of a variable. i.e. if $s = a + bx$, $t = c + dy$, where a, b, c, d are constants, b and d both positive, then $r_{st} = r_{xy}$. The product moment correlation coefficient for a set of data (x, y) is denoted by r. Express, in terms of r, the product moment correlation coefficient when all values of x are increased by 5. <u>Solution</u> There is no change to the product moment correlation coefficient when all values of x are increased by 5.
3	Draw scatter diagrams to represent situation where the product moment correlation coefficient for bivariate data is -1 , 0 or 1	Draw separate scatter diagrams, each with 8 points, all in the first quadrant, which represent the situation where the product moment correlation coefficient between the variables x and y is (i) -1, (ii) 0, (iii) Between 0.5 and 0.9.

No.	Skills	Examples of questions involving the skills																
		<u>Solution</u> When $r = -1$  When $r = 0$  When r is between 0.5 and 0.9 																
4	Draw scatter diagram for transformed data.	<table border="1"><tr><td>d</td><td>14588</td><td>8954</td><td>24 030</td><td>37 551</td><td>45 452</td><td>30 100</td><td>5055</td></tr><tr><td>p</td><td>12 450</td><td>15 200</td><td>11 900</td><td>10 700</td><td>10 600</td><td>10 900</td><td>22 500</td></tr></table> Using a reciprocal transformation, sketch a scatter diagram for p against $\frac{10\,000}{d}$. Draw, by eye, a line of best fit on your scatter diagram.	d	14588	8954	24 030	37 551	45 452	30 100	5055	p	12 450	15 200	11 900	10 700	10 600	10 900	22 500
d	14588	8954	24 030	37 551	45 452	30 100	5055											
p	12 450	15 200	11 900	10 700	10 600	10 900	22 500											

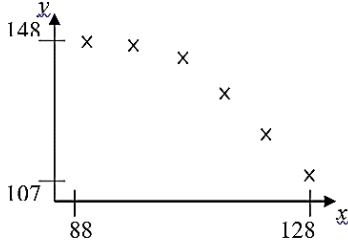
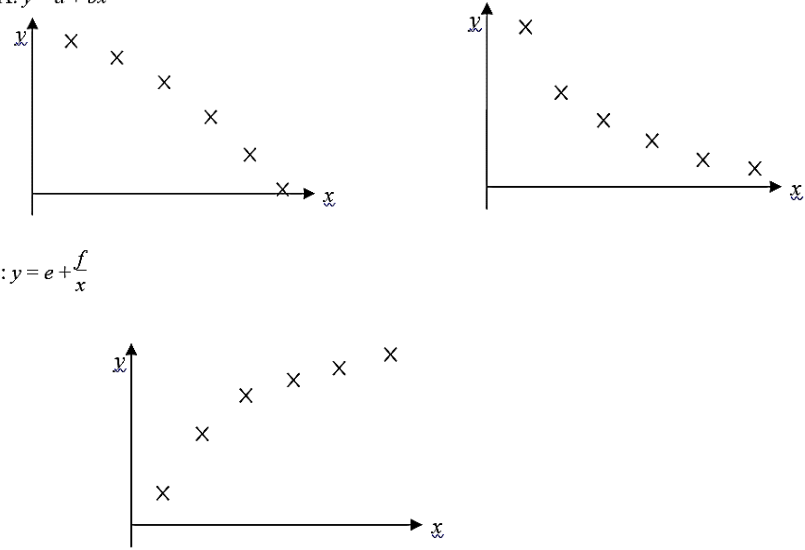
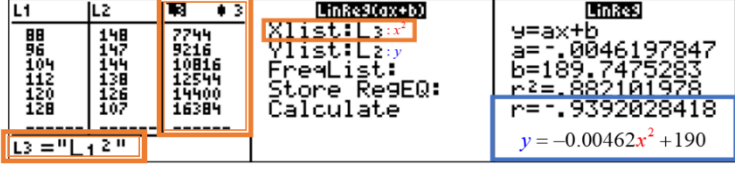
No.	Skills	Examples of questions involving the skills
		Solution  
5	Understand concepts of linear regression and method of least squares to find the equations of the regression lines	• Using method of least squares to find equations of regression lines The following summarizes the data from 10 sets of lengths(x) and breadths(y) in mm: $\sum x = 1782, \sum y = 1483, \sum x^2 = 318086,$ $\sum y^2 = 220257, \sum xy = 264582$ Find the equation of the regression line of y on x and that of x on y. Solution $b = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\left(\sum x^2 - \frac{(\sum x)^2}{n} \right)} = 0.58358; \bar{x} = 178.2; \bar{y} = 148.3$ $y - \bar{y} = b(x - \bar{x}) \Rightarrow y = 0.584x + 44.3 \text{ (to 3 s.f.)}$

No.	Skills	Examples of questions involving the skills																
		$d = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\left(\sum y^2 - \frac{(\sum y)^2}{n} \right)} = 0.94910$$x - \bar{x} = d(y - \bar{y}) \Rightarrow x = 0.949y + 37.4 \text{ (to 3 s.f.)}$ Use of $\bar{x}$ and $\bar{y}$ to solve for unknown/missing data A random sample of eight pairs of values of x and y is used to obtain the following equations of the regression lines of y on x and x on y respectively. $y = -\frac{7}{10}x + \frac{151}{10}, \quad x = -\frac{7}{6}y + 20$ Seven of the pairs of data are given in the table. <table><tr><td>x</td><td>10</td><td>11</td><td>12</td><td>11</td><td>17</td><td>14</td><td>19</td></tr><tr><td>y</td><td>9</td><td>8</td><td>7</td><td>6</td><td>5</td><td>4</td><td>1</td></tr></table> Find the eighth pair of values of x and y. <u>Solution</u> $\bar{y} = -\frac{7}{10}\bar{x} + \frac{151}{10} \text{ ---(1) and } \bar{x} = -\frac{7}{6}\bar{y} + 20 \text{ ---(2)}$	x	10	11	12	11	17	14	19	y	9	8	7	6	5	4	1
x	10	11	12	11	17	14	19											
y	9	8	7	6	5	4	1											

No.	Skills	Examples of questions involving the skills																																	
		Solving (1) & (2) gives $\bar{x} = 13, \bar{y} = 6$ $\bar{x} = \frac{\sum x}{n} = \frac{94 + x_8}{8} = 13 \Rightarrow x_8 = 10$ $\bar{y} = \frac{\sum y}{n} = \frac{40 + y_8}{8} = 6 \Rightarrow y_8 = 8$ The eighth pair of values is $(10, 8)$. • Explain the meaning of least square Explain why the residuals are squared, and why this is referred to as the ‘method of least squares’. Solution: Sum of the residuals may be negative, thus summing them up may give an erroneous results. The smallest squares of these residuals will give the best fit by calculation.																																	
6	Understand concepts of interpolation and extrapolation Use of appropriate regression lines to make prediction, estimate and explain how well the situation is modelled by the regression model	• When we do estimation within the given range of values of the data, it is known as interpolation. • When we do estimation outside the given range of values of the data, it is known as extrapolation. Values obtained by extrapolation may not be reliable since the relationship between the two variables may not follow the same linear model outside the range of the data. In a physical education class, the number of push-ups (x) and sit-ups (y) done by a sample of ten randomly chosen students were recorded in the table below. <table><tr><td>Student</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>x</td><td>27</td><td>22</td><td>15</td><td>35</td><td>30</td><td>52</td><td>35</td><td>55</td><td>40</td><td>40</td></tr><tr><td>y</td><td>30</td><td>26</td><td>25</td><td>42</td><td>38</td><td>40</td><td>32</td><td>54</td><td>50</td><td>43</td></tr></table> (iii) Find the equation of the regression line of x on y.	Student	1	2	3	4	5	6	7	8	9	10	x	27	22	15	35	30	52	35	55	40	40	y	30	26	25	42	38	40	32	54	50	43
Student	1	2	3	4	5	6	7	8	9	10																									
x	27	22	15	35	30	52	35	55	40	40																									
y	30	26	25	42	38	40	32	54	50	43																									

No.	Skills	Examples of questions involving the skills
		(iv) Use a suitable line to predict the number of sit-ups a student can do when he can do 50 push-ups. Give a reason if the predicted value is reliable. (v) Give a reason whether it is reliable to use the equation in (i) to predict the number of sit-ups when 60 push-ups are done. Solution (iii) The GC can be used directly to find the equation of the regression line of y on x as well as the equation of the regression line of x on y.    From GC, the equation of the regression line of x on y is $x = 1.07y - 5.60$ (to 3 s.f.). (iv) Since there is no clear independent variable in this case and we need to estimate y for a given value of x, we use the regression line of y on x.  

No.	Skills	Examples of questions involving the skills														
		From GC, the equation of the regression line of y on x is $y = 0.658x + 14.9 \text{ (to 3 s.f.)}.$ [Note that for this question, $15 \leq x \leq 55$ and $25 \leq y \leq 54$.] When $x = 50$, $y = 0.657885 (50) + 14.908225 = 47.8$ (to 3 s.f.). He can do 47.8 sit ups $\approx$ 48 sit ups. Since $x = 50$ is within the data range of x and $r = 0.839$ (to 3 s.f.) is close to 1, then there is a strong positive linear correlation between x and y . Therefore the estimated value of y obtained should be reliable. (iv) Since $x = 60$ is outside the data range, it is not reliable to do extrapolation and hence the estimated value is unreliable.														
7	Use of suitable transformations such as square, reciprocal or a logarithmic transformation to achieve linearity.	A motoring website gives the following information about the distance travelled, y km, by a certain type of car at different speeds, x kmh⁻¹ , on a fixed amount of fuel. <table><tr><td>Speed, x</td><td>88</td><td>96</td><td>104</td><td>112</td><td>120</td><td>128</td></tr><tr><td>Distance, y</td><td>148</td><td>147</td><td>144</td><td>138</td><td>126</td><td>107</td></tr></table> (ii) Draw the scatter diagram for these values, labelling the axes. (iii) Explain which of the three cases (A) $y = a + bx^2$, where a is positive and b is negative. (B) $y = c + d \ln x$, where c is positive and d is negative. (C) $y = e + \frac{f}{x}$, where e is positive and f is negative. is the most appropriate for modelling these values, and calculate the product moment correlation coefficient for this case and find the equation of a suitable regression line. <u>Solution</u> (ii)	Speed, x	88	96	104	112	120	128	Distance, y	148	147	144	138	126	107
Speed, x	88	96	104	112	120	128										
Distance, y	148	147	144	138	126	107										

No.	Skills	Examples of questions involving the skills
		 (iii) A: $y = a + bx^2$ B: $y = c + d \ln x$ C: $y = e + \frac{f}{x}$  By comparing the shape of the scatter diagram with the 3 different models, observe that the data points seem to lie close to a curve with negative gradient and concave downwards. Therefore we conclude that model A is the most appropriate. Note: Should all the proposed models have similar shape of curve, then compare the product moment correlation coefficient. Then use GC to calculate the product moment correlation coefficient for this case and find the equation of a suitable regression line. 

Supplementary Exercise

1. Eight pairs of observations on the variables x and y are given below:

x	16	13	16	15	16	12	17	15
y	7	17	12	11	18	10	17	12

- Find the regression line y on x .
- Find the regression line x on y .
- Find the product moment correlation coefficient for the data.
- Use an appropriate regression line to estimate the integer value of x if the value of y is correctly determined as 14. Explain why this is not a good estimation.
- Find $(\bar{x}, \bar{y})$ using (i) and (ii).

2. [CJC2019/Prelim/II/7]

An experiment is being carried out to study the correlation between the solubility of sugar in water and the temperature of the water. The amount of sucrose x (in grams) dissolved in 100 ml of water for different water temperatures T (in degree Celsius) is recorded. The results are shown in the table. Unfortunately, one of the values of x was accidentally deleted from the records later on and it is indicated by k as shown below.

T	0	20	40	60	80	100
x	179	204	241	k	363	487

It is given that the equation of the regression line of x on T is $x = 2.94857T + 146.238$. Show that $k = 288$. [2]

- Draw a scatter diagram to illustrate the data, labelling the axes clearly. [1]

The value of the product moment correlation coefficient between x and T is 0.959. It is thought that a model given by the formulae $\ln x = a + bT$ may also be a suitable fit to the data where a and b are constants.

- Calculate least square estimates of a and b and find the value of the product moment correlation coefficient between T and $\ln x$. [2]
- Use your answers to parts (i) and (ii) to explain which of $x = 2.94857T + 146.238$ or $\ln x = a + bT$ is the better model. [2]
- Hence, predict the value of T for which $x = 300$. Comment on the reliability of your prediction. [2]

3. [ASRJC2019/Prelim/II/5]

Greenhouse gases generated by Singapore comes mainly from the burning of fossil fuels to generate energy for industries, buildings, households and transportation.

The following table* shows the total greenhouse gas emissions, X (kilotonnes of CO_2), and the total deaths in Singapore, Y from 2007 to 2016.

Year	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
Total Greenhouse Gas Emissions (X)	42613	41606	43100	47063	49930	48712	50299	50908	51896	51519
Total Deaths (Y)	17140	17222	17101	17610	18027	18481	18938	19393	19862	20017

*Extracted from www.singstat.gov.sg

- (i) Draw a scatter diagram for these values, labelling the axes and state the product moment correlation between X and Y . Use your diagram to explain why the best model for the relationship between X and Y may not be given by $Y = aX + b$, where a and b are constants. [3]
- (ii) Comment on the correctness of the following statement.
 “Since there exists a high positive linear correlation between total greenhouse gas emission and total deaths, the deaths are a result of the greenhouse gas emissions.” [1]
- (iii) Explain clearly which of the following equations, where a and b are constants, provides the more accurate model of the relationship between X and Y .
 (A) $Y = aX^4 + b$
 (B) $\lg Y = aX + b$ [3]

The Government has pledged that Singapore’s greenhouse gas emissions will peak around 2030 at the equivalent of 65 million tonnes of carbon dioxide.

- (iv) Using the model you have chosen in part (iii), write down the equation for the relationship, giving the regression coefficients correct to 5 significant figures. Hence find the estimated total deaths in the year 2030 if Singapore’s pledge is fulfilled and comment on the reliability of the estimate. [3]

4. [HCI2019/Prelim/II/6]

Miss Tan carried out an investigation on whether there is a correlation between the amount of time spent on social media and exam scores. The average amount of time spent per month on social media, x hours, and the final exam score, y marks, of 6 randomly selected students from HCI were recorded. The data is shown below.

x	80	84	70	74	58	48
y	44	40	49	45	58	82

- (i) Draw a scatter diagram to illustrate the data. [2]
- (ii) It is found that the inclusion of a 7th point (x_7, y_7) will not affect the product moment correlation coefficient for the data. Find a possible point (x_7, y_7) . [1]

Omit the 7th point (x_7, y_7) for the rest of this question.

- (iii) State, with reason, which of the following equations, where a and b are constants, provides the most appropriate model for the relationship between x and y .

(A) $y = a + bx^2$,

(B) $e^y = ax^b$,

(C) $y = a + b\sqrt{x}$.

[3]

- (iv) Using the model chosen in part (iii), estimate the score of a student who spent an average of 60 hours per month on social media, giving your answer correct to the nearest whole number. [2]
- (v) Sam spends an average of 4 hours a day on social media. Assuming a 30-day month, suggest whether it is still reasonable to use the model in part (iii) to estimate his score. [1]

5. [CJC Prelim 9758/2017/02/Q6]

The table below shows the petrol mileage, y km/L and the weight, x kg in thousands for various car models in the year 1995.

x	3.5	3	2.75	2.5	2.25	2	1.75	1.5	1.25
y	7.5	8.0	8.5	8.7	10.0	k	13.5	16.8	18.0

- (i) The equation of the regression line of y on x is $y = 22.51355 - 4.908387x$. Show that $k = 11.0$. [2]
- (ii) Draw a scatter diagram to illustrate the data. [1]

- (iii) With reference to the scatter diagram and context of the question, explain why model (C) below is the most appropriate for modelling the data as compared to the other 2 models.
- (A) $y = a + bx$, where a is positive and b is negative,
 (B) $y = a + b \ln x$, where a is positive and b is negative,
 (C) $y = a + \frac{b}{x}$, where a and b are positive.
- (iv) Calculate the least squares estimates of a and b for model (C). [1]
- (v) Predict the weight of the car if the petrol mileage is 12 km/L. Comment on the reliability of your prediction. [1]
- (vi) Suppose there was an error in recording the y values and all the y values must be increased by a constant M km/L, state any change you would expect in the values of
- (a) $\bar{y}$, [1]
 (b) standard deviation of y [1]
 (c) the correlation coefficient. [1]

6. [JJC Prelim 9758/2017/02/Q10]

- (a) Traffic engineers are studying the correlation between traffic flow on a busy main road and air pollution at a nearby air quality monitoring station. Traffic flow, x , is recorded automatically by sensors and reported each hour as the average flow in vehicles per hour for the preceding hour. The air quality monitoring station provides, each hour, an overall pollution reading, y , in a suitable unit (higher readings indicate more pollution). Data for a random sample of 8 hours are as follows.

Traffic flow, x	1796	1918	2120	2315	2368	2420	2588	2850
Pollution reading, y	1.0	2.2	3.5	4.2	4.3	4.5	4.9	5.5

- (i) Draw the scatter diagram for these values, labelling the axes. [2]

It is thought that the pollution y can be modelled by one of the formulae

$$y = a + bx \qquad y^2 = c + dx$$

where a , b , c and d are constants.

- (ii) Find the value of the product moment correlation coefficient between
- (a) x and y ,
 (b) x and y^2 . [2]
- (iii) Use your answers to parts (i) and (ii) to explain which of $y = a + bx$ or $y^2 = c + dx$ is the better model. [2]
- (iv) It is required to estimate the value of y for which $x = 2000$. Find the equation of a suitable regression line, and use it to find the required estimate. [2]

- (v) The local newspaper carries a headline “Heavy traffic causes air pollution”. Comment briefly on the validity of this headline in the light of your results. [1]
- (b) The diagram below shows an old research paper that has been partially destroyed. The surviving part of the paper contains incomplete information about some bivariate data from an experiment. Calculate the missing constant at the end of the equation of the second regression line. [3]

The mean of x is 4.4. The
 The equation of the regression line of y on x is $y = 2.5x + 3.8$.
 The equation of the regression line of x on y is $x = 1.5y -$

7. [RI JC2 Prelim 9758/2019/02/Q10]

- (a) With the aid of suitable diagrams, describe the differences between the least square linear regression line of y on x and that of x on y . [2]
- (b) The government of the Dragon Island Country is doing a study on its population growth in order to implement suitable policies to support the aging population of the country. The population sizes, y millions in x years after Year 2000, are as follows.

x	9	10	11	12	14	15	16	17	18
Population size, y	5.05	5.21	5.32	5.41	5.51	5.56	5.60	5.65	5.67

- (i) Draw a scatter diagram of these values, labelling the axes clearly. Use your diagram to explain whether the relationship between x and y is likely to be well modelled by an equation of the form $y = ax + b$, where a and b are constants. [2]
- (ii) Find, correct to 6 decimal places, the value of the product moment correlation coefficient between
- (a) $\ln x$ and y ,
- (b) x^2 and y . [2]
- (iii) Use your answers to part (ii) to explain which of $y = a + b \ln x$ or $y = a + bx^2$ is the better model. [1]
- (iv) It is required to estimate the year in which the population size first exceed 6.5 millions. Use the model that you identified in part (iii) to find the equation of a suitable regression line, and use your equation to find the required estimate.

Comment on the reliability of this estimate. [3]

- (v) As the population size in Year 2013 is not available, a government statistician uses the regression line in part (iv) to estimate the population size in 2013. Find this estimate. [1]
- (vi) It was later found that in Year 2013, the population size was in fact 5.31 millions. Comment on this figure with reference to the estimate found in part (v), providing a possible reason in context. [1]

8. [VJC2019/Prelim/II/7]

Seven members of the school cross country team undergo a new training programme to improve their fitness. During a particular session, each of them has to complete a 200 metre run and to achieve as many push-ups as possible in one minute. The times taken for the 200 metre run, t seconds, together with the number of push-ups each runner achieves, n , are shown in the table.

Student	A	B	C	D	E	F	G
t	38.3	42.1	35.1	40.1	32.0	31.6	41.0
n	44	35	48	42	49	49	40

- (i) Draw a scatter diagram to illustrate the data, labelling the axes. [1]
- (ii) Explain using your scatter diagram why the linear model $n = at + b$ would not be appropriate. [1]

It is thought that the relationship between n and t can be modelled by one of the formulae

$$n = c(t - 30)^2 + d \quad \text{or} \quad n = e(t - 30)^3 + f$$

where c , d , e and f are constants.

- (iii) The product moment correlation coefficient between n and $(t - 30)^2$ is -0.980 , correct to 3 decimal places. Determine, with a reason, which of the 2 models is more appropriate. [2]
- (iv) It is known that student H is able to do 48 push-ups in one minute. It is required to estimate student H's timing for the 200 metre run. Find the equation of a suitable regression line, and use it to find the required estimate. Comment on the reliability of this estimate. [5]

9. [SAJC 2023/Prelim/II/10]

Tom records the maximum speeds of an athlete running in a temperature controlled environment. The table below shows the maximum speed, y metres per second, recorded for various specific temperatures, x degree Celsius.

x	6	11	16	21	26
y	2.96	2.94	2.85	2.71	2.52

- (i) Tom decides to model the data using the line $y = -0.02x + 3.1$.

- (a) On the grid opposite
- draw a scatter diagram of the data, and
 - draw the line $y = -0.02x + 3.1$. [3]
- (b) For a line of best fit $y = f(x)$, the residual for a point (a, b) plotted on the scatter diagram is the vertical distance between $(a, f(a))$ and (a, b) . Mark the residual for each point on your diagram. [1]
- (c) Calculate the sum of the squares of the residuals for Tom's line. [1]
- (d) Explain why, in general, the sum of the squares of the residuals rather than the sum of the residuals is used. [1]
- (ii) State the value of the product moment correlation coefficient. [1]
- (iii) Use your calculator to find the equation of a suitable regression line to estimate the temperature for a given maximum speed. Hence estimate the temperature when the maximum speed is 2.48 metres/second. Explain whether you would expect this value to be reliable. [4]
- (iv) Explain whether the product moment correlation coefficient would differ if Tom converted the speed to kilometres per hour. [1]

10. [RVHS2018/CT2/I/8]

- (a) Draw separate scatter diagrams, each with 8 points, in the first quadrant, which represent the situation where the product moment correlation coefficient between variables x and y is
- (i) 0.999 [1]
 - (ii) 0 [1]
- (b) The values of 8 observations of bivariate data, x and y are as shown in the table below.

x	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5
y	28.4	26.5	24.1	21.3	18.2	14.5	9.9	5.1

- (i) Find the value of the product moment correlation coefficient for the data. Explain why this value does not necessarily imply that the best model for the relationship between x and y is $y = a + bx$. [2]
- (ii) Determine, with reason, whether $y = a + bx$ or $y = c + dx^2$ is a better model for the above set of data. [1]
- (iii) By using the appropriate regression line, estimate the value of y when $x = 3.2$ and comment on its reliability. [3]

11. [NJC2018/Prelim/II/11]

- (i) Sketch a scatter diagram, that might be expected when x and y are related approximately, for each of the cases (A) and (B) below. In each case your diagram should include 6 points, approximately equally spaced with respect to x , and with all x - and y - values positive.

(A) $y = a + bx^2$, where $a, b \in \mathbb{R}^+$

(B) $y = c + \frac{d}{x}$, where $c \in \mathbb{R}^+$ and $d \in \mathbb{R}^-$. [2]

Obesity is becoming increasingly prevalent across the globe. To investigate the effects of obesity on one's health, a study was conducted to determine if the blood pressure of adults aged between 40 and 50 years old is dependent on their Body Mass Index (BMI). Data from six patients in this age-group from a hospital are collected. Their BMI, m kg/m², and systolic blood pressure, s , in mmHg, are as follows.

m	22	27	31	36	40	44
s	120	150	168	172	179	183

- (ii) Sketch the scatter diagram for these values, labelling the axes clearly. [2]
- (iii) With reference to your scatter diagram in part (ii), explain why model (B) in part (i) is more appropriate than model (A) for modelling these values and calculate the product moment correlation coefficient for model (B). [2]
- (iv) Find the equation of the least-squares regression line of s on $\frac{1}{m}$ and use it to estimate the BMI of another patient (of a similar age profile) whose systolic blood pressure is 110 mmHg. Comment on the reliability of your estimate. [3]
- (v) Explain why the regression line of $\frac{1}{m}$ on s should not be used for your calculations in part (iv). [1]
- (vi) State, in context, a limitation of using the regression equation in part (iv) to estimate the systolic blood pressure of *other* people with BMI within the range $22 < m < 44$. [1]
- (vii) Suppose a new data pair $(\bar{m}, \bar{s})$ is added to the table above, where $\bar{m}$ and $\bar{s}$ are the mean BMI (in kg/m²) and the mean systolic blood pressure (in mmHg) of the adults in the study respectively, based on the data above. Without any calculations, explain whether the equation of the regression line you have obtained in part (iv) would change. [1]

12. [ACJC Prelim 9758/2024/02/Q10]

A female social media influencer is analysing the performance of her past videos posted online. She wants to understand the relationship between the number of video views, v , and the number of followers gained, f , from her past video posts. The data from her earliest 9 videos is given in the table below.

v	1000	5000	8000	10 000	20 000	30 000	40 000	50 000	60 000
f	15	163	10	278	389	456	492	541	560

- (a) Due to a technical issue, one of the 9 videos had no audio which affected the number of followers gained. Draw a scatter diagram of these 9 data points and circle the data point that likely represents the video with no audio. [2]

The female social media influencer decides to **exclude** the data point that represents the video with no audio from her analysis. For parts (b), (c) and (d) of this question, you should **exclude** this data point.

- (b) Use your scatter diagram to explain with reasons the conclusion that the influencer should reach regarding the relationship between f and v . [1]
- (c) By referring to the scatter diagram and calculating the relevant product moment correlation coefficients, determine whether the relationship between f and v is modelled better by $f = a + bv$ or $f = a + b \ln v$, where a and b are constants.

Explain how you decide which model is better and state the equation of the least squares regression line in this case, giving your answer to 3 decimal places. [5]

- (d) Use the appropriate least squares regression line in (c) to estimate the number of followers gained when the number of video views is 100 000. Comment on the reliability of this estimate. [2]
- (e) Explain why in the 'method of least squares', the distances which are used in finding the least squares regression line are squared. [1]

A male social media influencer also decides to analyse the performance of his past videos posted online and found that the sum of the squares of the distances which are used in finding the least squares regression line is zero.

- (f) What can you deduce about the data points of this male influencer? [1]

13. A random sample of eight pairs of values of x and y is used to obtain the following equations of the regression lines of y on x and x on y respectively.

$$y = -\frac{7}{10}x + \frac{151}{10}, \quad x = -\frac{7}{6}y + 20$$

Seven of the pairs of data are given in the table.

x	10	11	12	11	17	14	19
y	9	8	7	6	5	4	1

Find the eighth pair of values of x and y .

Draw a scatter plot for the above data and find the value of the product moment correlation coefficient between x and y . Comment on the suitability of the use of a linear model for the variables x and y .

Let Y be the value obtained by substituting a sample value of x into the equation of the regression line of y on x . Evaluate Y for each of the eight values of x and find the value of $\sum (y - Y)^2$.

For each of the eight sample values of x , Y' is given by $Y' = a + bx$, where a and b are any constants. What can you say about the value of $\sum (y - Y')^2$?

14. [YJC2018/Prelim/II/9]

A new Internet service provider in the market, Y-Fai, decides to investigate the effect of the distance from its router on its wifi signal. The most convenient way to express wifi signal strength is by using dBm, which stands for decibels relative to a milliwatt. The signal strength measured in dBm is negative in value and a value closer to 0 signifies a stronger wifi signal. An employee measures the signal strength (y dBm) at various fixed distances away from its router (x m) as follows.

x	1	1.5	2	2.5	3	10	15
y	-30	-60	-100	-76	-79	-83	-88

- (i) Draw a scatter diagram for these values. On your diagram, circle the data point that seems to be unexpected and suggest a possible reason for the anomaly. [2]

For parts (ii) and (iii) of this question, you should **exclude** the anomaly.

- (ii) Explain from your scatter diagram why the relationship between x and y should not be modelled by an equation of the form $y = ax + b$. [1]
- (iii) Which of the formulae $y = \frac{c}{x} + d$ and $y = \frac{e}{x^2} + f$, where c , d , e and f are constants, is the better model for the relationship between y and x ? Explain fully how you decided, and find the constants for the better formula. [4]

- (iv) Use the formula you chose from part (iii) to estimate the signal strength when the distance away from its router is 5 m. Explain why you would expect this estimate to be reliable. [2]

15. [ACJC JC2 Prelim 9758/2019/02/Q8]

A company that organizes live concerts believes that the popularity of an artiste affects his/her concert ticket sales. The popularity of an artiste can be measured by an index, x , such that $1 \leq x \leq 10$, where 10 indicates most popular and 1 indicates least popular. A study was conducted over six months to investigate the relationship between the popularity index of eight artistes and their concert ticket sales. The results are summarised in the following table.

Artiste	A	B	C	D	E	F	G	H
Popularity Index, x	1.2	2.0	2.7	3.8	4.8	5.6	6.9	8.0
Concert Ticket Sales (hundreds of thousands), y	2.2	4.5	5.8	7.3	7.4	9.0	9.9	10.8

- (i) Draw a scatter diagram for these values, labelling the axes. [1]
 (ii) It is thought that concert ticket sales y can be modelled by one of the formulae

$$y = ax^2 + b \quad \text{or} \quad y = c \ln x + d,$$

where a , b , c and d are positive constants.

Use your diagram in (i) to explain which of the two is a more appropriate model, and calculate its product moment correlation coefficient, correct to 4 decimal places.

- [2]
 (iii) The data for a particular artiste appears to be recorded wrongly. Indicate the corresponding point on your diagram by labeling it P . Find the equation of the least squares regression line for the remaining points using the model that you have chosen in (ii). [2]

Answers:

- (i) $y = 0.35x + 7.75$ (ii) $x = 0.0648y + 14.2$ (iii) 0.151 (iv) 15 (v) (15, 13)
- (ii) $a = 5.13$; $b = 0.00987$; $r = 0.990$ (iv) $T = 58.2$
- (i) $r = 0.905$ (3 sf)
 (iv) $Y = 6.4326 \times 10^{-16} X^4 + 14911$; $Y = 26400$ (3 sf)
 (v) The estimate is not reliable as extrapolation is carried out where the linear correlation outside the range of X , i.e. [41606, 51896] may not be valid.
- (ii) $(\bar{x}, \bar{y}) = (69, 53)$ (iii) Model B is the most accurate model
 (iv) 62 marks (v) unreasonable
- (iv) $a = 0.257$, $b = 22.8$ (v) 1860kg
 (vi)(a) $\bar{y}$ will be increased by M . (b) remain unchanged.
 (c) remain unchanged.(i)
- (a) (ii) (a) 0.959, (b) 0.995 (iii) $y^2 = c + dx$ is the better model since

- (iv) $y^2 = 0.0279x - 48.0$, $y = 2.79$ when $x = 2000$.
 (v) May not be valid as correlation does not necessarily imply causation.
 (b) 17.8
7. (b)(ii) 0.986554, 0.945806 (b)(iv) $y = 3.24 + 0.854 \ln x$, 2045 (b)(v) 5.43 millions
8. (i) $n = e(t - 30)^3 + f$ is the more appropriate model of the two.
 (iv) $(t - 30)^3 = -132.12n + 6487.1$; $t = 35.3$
9. (i)(c) 0.0134, (ii) -0.958, (iii) $y = -0.0222x + 3.15$; 30.2 degree Celsius.
10. (b)(i) $r = -0.988$
 (b)(ii) $y = c + dx^2$ is a better model
 (b)(iii) $y = 16.9$
11. (iii) $y = c + \frac{d}{x}$, is more appropriate; $r = -0.982$
 (iv) $m \approx 19.706$
12. (c) $f = -976.691 + 138.580 \ln v$ (3 d.p.) (d) 619
13. The eighth pair of values is (10, 8). $r = -0.904$. $\sum (y - Y)^2 = 8.8$
14. (i) There could be a wifi disruption.
 (ii) The data points do not seem to lie on a straight line.
 (iii) $y = \frac{e}{x^2} + f$ is the better model; $e = 55.8$ (3 s.f.); $f = -85.4$ (3 s.f.)
 (iv) The required signal strength is -83.2 dBm
15. (ii) $r = 0.9927$ (4dp) (iii) $y = 1.38 + 4.46 \ln x$ (3 s.f.)
 (iv) $x = 4.42$ (3 s.f.)